



Pipeline Head Office
Maintenance Department

PLHO/MNT/MECH/05

April 20, 2018

Subject: Long term solution of water leakage problem of MP1 MPPL Abu Road

This is in continuation to note PLHO/MNT/MECH/05 dated 09.03.2018 regarding long term solution of water leakage from cylinder block of MP1 MPPL Abu Road. Vide said note PL HO maintenance proposed for replacement of MP1 MPPL Abu Road block with block of decommissioned old G series engine of MP2 SMPL Kot.

Subsequently, vide e-mail dated 21.03.2018 WRPL was requested to carry out further detailed analysis & assessment for execution and share detailed planning and timeline in this regard with this office.

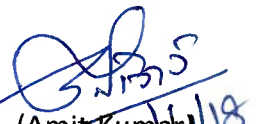
WRPL has worked out the detail plan and communicated vide mail dated 16.04.2018. Maintenance plan, activities and time line for the subject job has been reviewed, considering the planned maintenance & availability of spares. Accordingly, subject work is planned after September 2018. At the same time requirement of Allen engine spares were also considered in the LTSA proposal for the year 2018-19.

In view of the above, In-principal approval may please be accorded for permanent solution of water leakage problem of MP1 MPPL Abu Road. Upon approval, it may please be forwarded to WRPL for necessary planning and action.

Submitted please.

1. Entablature of MP, (MPPL Abu Road) is planned for change in Oct/Nov 2018 as per Region plan.
2. Engine downtime will be 50-60 days.

GM (Maint)


(Amit Kumar)
CMNM, PLHO


23-4-18

CGM (M&I)

Please see that during the proposed work other engines of MPPL, Abu Road are fully operational.

- Test and inspection of the block at Kot shall be thorough, ..

- A team may be formed to execute the work.

Proposal is approved

CMNM (AK)

GM (M&I)




26-4-2018

Action Plan for Engine block replacement of MP-1 MPPL Abu Road											
S. No.	Activity	Month of FY 2018-19						Remarks			
		May'18	June'18	July'18	Aug'18	Sep'18	Oct'18	Nov'18	Dec'18		
1	Dismantling of Engine Block of MP-2 SMPL KOT									MP-2 Kot entablature is planned to be used	
2	Metallurgical tests i.e MPI/ DP/UT of Engine Block of MP-2 SMPL KOT										
3	Shifting of Engine Block of MP-2 SMPL KOT to SMPLABUROAD										
4	Painting of Engine Block of MP-2 SMPL KOT										
5	Arrangement of Critical spares i.e. Main bearings/ Liners/Pistons/ Big end bearings etc through existing stock /LTSA 2018-19									LTSA Spares will be required therefore considered after Sep'18. PLHO Maint also requested accordingly.	
6	Arrangement of Consumable Spares										
7	Preparation and Award of work for Manpower Assistance										
8	Preparation and Award of work for Hiring of crane services										
9	Preparation and Award of work for Miscellaneous repair/ job work including re-dowelling job etc										
10	Preparation and Award of work for Lining up M/s Allen Diesel during re-dowelling of block/ gear timing/ Final commissioning									Services of M/s Allen diesels considered for critical jobs	
11	Replacement of engine block including 18000 Hrs major maintenance & trial run										
12	Stabilization , Performance analysis & Preparation of maintenance reports										

Kumar, Amit (कुमार, अमित)

From: Kumar, Amit ()
Sent: Saturday, April 21, 2018 2:45 PM
To: Kumar, Amit ()
Subject: FW: Long term solution of MP1 MPPL Aburoad
Attachments: Copy of Action Plan for Engine block replacement of MP-1 MPPL Abu Road.xlsx;
Action Plan for Engine block replacement of MP-1 MPPL Abu Road.pdf

From: Gupta, Manoj (गुप्ता, मनोज)
Sent: Monday, April 16, 2018 3:59 PM
To: Kumar, Amit (कुमार, अमित)
Cc: Banerjee, D. K (बनर्जी, डी. के.); GUPTA, S.S. (PLHO) (एस.एस. गुप्ता); PANDEY S P (एस. पी. पांडे); SHENAI, M. R.(एम.आर. शनई); PRAJAPATI, RAKESH (राकेश प्रजापति)
Subject: FW: Long term solution of MP1 MPPL Aburoad

Dear Amit,

As desired, action plan and timeline related to Long term solution of MP1 MPPL AbuRoad is placed below.
It is also requested to kindly review the same and inform in case of any observations etc.

Action Plan for Engine block replacement of MP-1 MPPL Abu Road										
S. No.	Activity	Month of FY 2018-19								Remarks
		May'18	June'18	July'18	Aug'18	Sep'18	Oct'18	Nov'18	Dec'18	
1	Dismantling of Engine Block of MP-2 SMPPL KOT									MP-2 Kot entablature is planned to be used
2	Metallurgical tests i.e. MPI/ DP/UT of Engine Block of MP-2 SMPPL KOT									
3	Shifting of Engine Block of MP-2 SMPPL KOT to SMPPL ABUROAD									
4	Painting of Engine Block of MP-2 SMPPL KOT									
5	Arrangement of Critical spares i.e. Main bearings/ Liners/Pistons/ Big end bearings etc through existing stock /LTSA 2018-19									LTSA Spares will be required therefore considered after Sep'18. PLHO Maint. also requested accordingly.
6	Arrangement of Consumable Spares									
7	Preparation and Award of work for Manpower Assistance									
8	Preparation and Award of work for Hiring of crane services									
9	Preparation and Award of work for Miscellaneous repair/ job work including re-dowelling job etc									
10	Preparation and Award of work for Lining up M/s Allen Diesel during re-dowelling of block/ gear timing/ Final commissioning									Services of M/s Allen diesels considered for critical jobs
11	Replacement of engine block including 16000 Hrs major maintenance & trial run									
12	Stabilization , Performance analysis & Preparation of maintenance reports									

सादर (Regards)

मनोज गुप्ता | Manoj Gupta

उप महाप्रबन्धक (अनुरक्षण) | DGM (Maintenance)

पश्चिमी क्षेत्र पाइपलाइन्स | Western Region Pipelines

From: Kumar, Amit (कुमार, अमित)

Sent: 09 April 2018 18:11

To: Gupta, Manoj (गुप्ता, मनोज)

Cc: GUPTA, S.S. (PLHO) (एस.एस. गुप्ता); Banerjee, D. K (बनर्जी, डी. के.); PANDEY S P (एस. पी. पांडे); SHENAI, M. R.(एम.आर. शनई); Kumar, Ankit (कुमार, अंकित)

Subject: RE: Long term solution of MP1 MPPL Aburoad

Importance: High

Dear Sir,

This is in reference to the trailing mail. It is requested to share the detailed planning, action plan and timeline related to Long term solution of MP1 MPPL Aburoad.

Regards

Amit Kumar
CMNM, PLHO

From: Kumar, Amit (कुमार, अमित)

Sent: Wednesday, March 21, 2018 4:48 PM

To: Gupta, Manoj (गुप्ता, मनोज)

Cc: GUPTA, S.S. (PLHO) (एस.एस. गुप्ता); banerjee@INDIANOIL.IN; PANDEY S P (एस. पी. पांडे); SHENAI, M. R.(एम.आर. शनई); Kumar, Ankit (कुमार, अंकित)

Subject: Long term solution of MP1 MPPL Aburoad

Importance: High

Dear Sir,

In MP1 MPPL Abu Road, water leakage problem was first reported in 2011 and subsequently in 2015 & 2017. Solutions, suggested by M/s Allen Diesel, were adopted on each occasion and were temporary leading to re-occurrence after time gap .

From the re-occurrence data it is clear that the problem is aggregating with passage of time - the time period between two water leakages has decreased.

In view of the above, a long lasting /permanent solution of this engine is essential.

The table indicating three plausible options, with the influencing factors (estimated) for permanent solution of said engine, on the basis of past experience of similar problems in Old G series Allen engines, is given below;

SN	Option No	Repair Methodology	Time of Non Availability	Total Duration of Execution	Total Cost
1	Option I	In situ repair of engine block (previously done in MP2 Sendra in 2013 & No water leakage since repair)	4 months	12-14 months	Rs 2.0 – 2.5 Cr
2	Option II	Replacement of block with spare block (previously done in MP2 Kot in 2009 & No water leakage since block replacement)	2 months	3-4 months	Rs 1.0 -1.2 Cr
3	Option III	Replacement of block with a new top flat design block (No new G sr engine block procured previously)	2 months	14-16 months	Rs 4.0 – 4.5 Cr

Option II seems more economical with implementation time also lesser.

Therefore PLHO proposes for replacement of MP1 MPPL Abu Road block with block of decommissioned old G series engine of MP2 SMPL Kot.

In view of the above, WRPL is requested to carry out further detailed analysis & assessment for execution, prepare detailed planning and timeline in this regard and share with this office.

सादर/ Regards

अमित कुमार / Amit Kumar

वरिष्ठ अनुरक्षण प्रबन्धक / Senior Maintenance Manager

इंडियन ऑयल कॉर्पोरेशन लिमिटेड/ Indian Oil Corporation Limited

पाइपलाइन्स प्रभाग (मुख्यालय) / Pipelines Division (Head Office)

मोबाइल/ Mobile : 9426706267

दूरभाष/ Telephone : 011-2448908



Pipeline Head Office
Maintenance Department

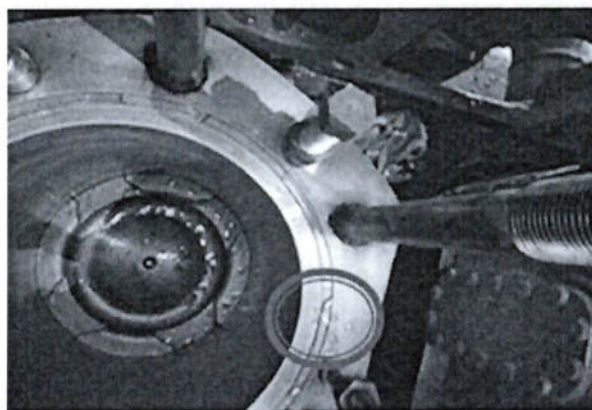
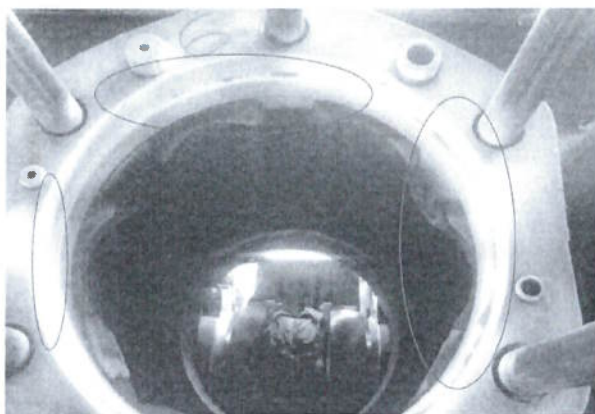
PLHO/MNT/MECH/05

March 8, 2018

Sub : Permanent solution to water leakage from cylinder block of MP1 MPPL, Aburoad

1. Problem

Continuous leakage of engine jacket cooling water is being observed in MP-1 at MPPL Abu Road from the area between liner collar and cylinder heads due to crack in the engine block in liner landing area. Operation team has to fill up the makeup water tank in every 3-4 hours during operation of engine.



2. Background of MP1 MPPL Abu Road

3 Nos of Allen S 37 G series engines were installed at Abu Road under KBPL pipeline project. Later on KBPL product pipeline was converted to MPPL crude oil pipeline. The details of engine are placed below.

Equipment	MP-1/ MP-2 / MP-3
Engine Serial Number	D6/50161-4/5/6
Make	M/s. WH ALLEN, UK
Type	8S37G HBC

On October 2011, water leakage was observed between engine block and Cylinder head from cylinder No. 6, 7 and 8 of MP1. Crankcase doors opened & no water leakage observed. Cylinder head also removed and No water found in the combustion chamber.

These were external leakages and were similar to those earlier observed in old G series Allen engines at Sidhpur (MP1 & MP2), Kot (MP2) and Sendra (MP1 & MP2). Since there was no internal leakage in engine, it was then decided to address the issue during major maintenance which was due in November, 2011.

During the 9000hrs schedule maintenance (CRH: 51807), started from 14.11.2011, detailed inspection was carried out and cracks were observed in the entablature (in all cylinders) at liner collar landing area. The matter was deliberated with M/s Allen Diesel and water transfer sleeve were replaced with new modified water transfer sleeves. **Flag - A.**

In November 2015, at CRH: 67290, after running of 15,843 Hrs, external water leakage was again observed from block of Cylinder head No. 8. New sleeves were fabricated of extended length and fitted in Cylinder #8 of engine. After replacement of sleeves, engine trial run was taken and no water leakage was observed. **Flag - B**

3. Recent issues in the Engine

Water leakage from engine block increases from April 2017 in cylinder No 7 & 8 and major maintenance of MP-1 was also due hence 12000 Hrs maintenance was carried out from 12th May to 7th July 2017 at CRH: 72858. M/s Rai Prexim, Gurgaon was also lined up for inspection of Cylinder block for cracks, checking condition of water transfer sleeves etc. MPI of engine block at liner landing area was carried out which revealed similar circumferential cracks which were reported in 2015. These cracks might have propagated this time further downward. Dimensional inspection of water transfer sleeves, crack and ovality checking also carried out. Thereafter, sleeves were re-fitted and trial run was taken.

During the load trial on 07th July 2017, external water leakage was observed from cylinder No. 3, 4, 7 & 8 however No water leakage observed from other cylinders. Water was coming out from liner landing area.

The cracks observed were likely to propagate which may increase the amount of water leakage however it is not expected that these cracks would lead to any catastrophic failure of the engine if the set continues to operate, therefore MLPU made available for operation with water leakage from cylinder No 3, 4, 7 & 8.

The problems anticipated in the engine while operating with the above mentioned water leakages are placed below:

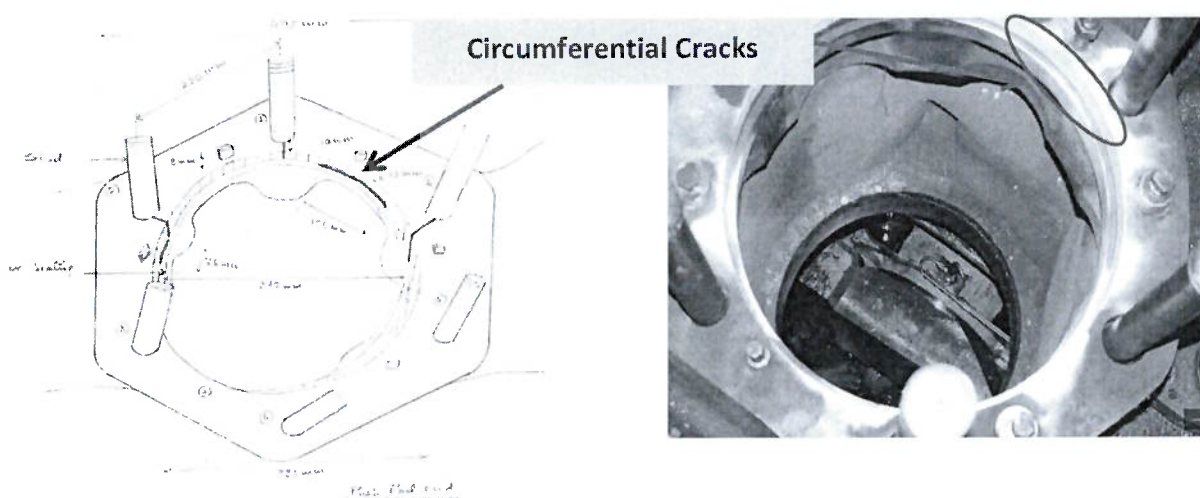
1. Jacket water will have to be topped up regularly to avoid water starvation in the engine.
2. Impossible maintaining correct dosage of water treatment chemicals.
3. Increase in consumption of jacket water & water treatment chemicals.
4. Failure of Cylinder head studs due to corrosion may also occur in long run.



5. Possibility of corrosion in J/w circuit and liner outer surface as correct dosage of water treatment chemical could not be maintained.
6. Failure of cylinder liner O ring & cylinder head O ring in case of water starvation.

4. History of the water leakage problem in G series Allen Engine

In 2003, the leakage of jacket water from the cylinder head-liner collar seating area was first noticed in cylinder No 1 & 2 of MP2 SMPL Sidhpur. DP test and soap bubble test was carried out and circumferential cracks were detected in liner landing area. Cracks were observed in the engine block close to liner collar seating area were cause of water leakage.



As a precaution cylinder No. 4 & 8 of this engine were also checked for peripheral cracks by plugging the water transfer pipes and introducing compressed air into the water transfer passage from the pocket inside. Dye penetrant test of the liner land area was also carried out. No leaks were detected in these cylinders.

The water was observed coming out of the cracks during return movement from engine block to cylinder head and was coming outside from the area between head & liner. Application of sealants on the cracked portion for containing the water leakage could not provide a solution to the problem.

This matter was deliberated with M/s Allen (OEM) with regard to MP2 Sidhpur and they have submitted a report No IOCL 2003 / 001 and issued service bulletin Number 016/4000 in 2003. As per service bulletin, Old G series engines, supplied at SMPL Kot, Sendra & Sidhpur, and Nine G series engines supplied under KBPL project i.e. installed at Kandla, Abu Road & Beawar were having liner flange dimensions which fits bit tighter in the block and therefore are susceptible to cracking of the block in cases of overload or overheating. **Flag - C.**

4.1. Reason attributable for crack in Engine block

The cracking of the cylinder block has most likely been caused by high hoop stresses in the liner seating area caused by an earlier incident of overload or overheating. At this time the cracks would have been instigated and it may have been a long period of propagation until the peripheral cracks reached the water transfer holes. On reaching the water transfer holes a leakage path for Jacket water is provided by these cracks.

This type of crack is generally caused by high hoop stresses which once initiated may propagate towards water transfer holes or to underneath the top portion of block to water jacket. However the possibility of crack propagation towards water transfer hole is more due to discontinuities in top portion of block because of six cylinder head stud holes and six water transfer hole that makes this portion weaker. These cracks provide the path for leakage of Jacket water.

Accordingly, Liner dimensions were changed to provide more clearance in the liner landing area. Inspection of liners fitted in these five old G series engines and nine original KBPL engines were carried out in the subsequent major maintenance and replaced accordingly.

4.2. Solutions to aforesaid problem

The problem was further analysed and consultation was done with OEM i.e. M/s Allen Diesel in 2003-04 regarding probable cause of water leakage and its best possible solutions. The OEM suggested following short term and long term solutions.

A. Short term Solution / Temporary solution

- Fitting of extended sleeve in water transfer holes and applying locktit 290 in all cracks.
- Filling with silicon base sealant around the neck of cylinder head stud to prevent water accumulation there.

B. Long term Solution

- **Repairing of cylinder block** - Machine 62 mm from the top of the cylinder to remove all the cracks and fit top liner inserts with cap screw bringing the liner landing back to nominal dimension.
- **Replacing the cylinder block**

4.3. Repair methodology adopted by IOCL

Over a period of time water leakage problem was observed in all the five old G series Allen Engines in SMPL i.e. MP1 & MP2 Sidhpur, MP2 Kot and MP1 & MP2 Sendra. Temporary

These cracks seems to have developed due to material inhomogeneity. NOT agreed by M/s Allen Diesel.

GS

solution i.e. extended water transfer sleeves were fitted in all the affected engines of SMPL. The above measure could provide a temporary solution to the problem, the water leakage problem again resurfaced. Simultaneously for permanent solution of the problem, steps were taken up in phased manner for these engines, details are placed below:

SN	Engine	Repair Methodology	Remarks
1	MP1 SMPL Sidhpur	Repairing of engine block carried out through third parties 1. By machining of top face and metal stitching of insert through M/s Rai Prexim, Gurgaon in 2004-05 by metal stitching. 2. By machining of top face and cast iron welding of insert through M/s Esdee Electro Mills in 2008-09.	Repairing of engine block was not completely successful as water leakage re-surfaced after 1 to 2 year of operation in both the cases. Heavy water leakage, temporary solutions carried out time and again. In every 2-3 hours water filling was required. MLPU de-commissioned in SMPL DBL
2	MP2 SMPL Sidhpur	Block replaced - Engine block was replaced with that of MP1 Kandla Engine in 2004.	No water leakage in this engine since block replacement MLPU is not in continuous operation after SMPL DBL however weekly trial is taken.
3	MP2 SMPL KOT	Block replaced - Block of burnt engine of Kot fire (New G series) was used with tailor made water transfer plate in 2009.	No water leakage in this engine since block replacement MLPU de-commissioned in SMPL DBL
4	MP1 Sendra	Short term solution only. Water transfer sleeve extension carried out. Only temporary solution adopted.	Water leakage problem persisted and water filling was required in every 2-3 hours. MLPU was decommissioned in SMPL DBL in 2016.
5	MP2 Sendra	Repairing of engine block by Allen Diesel - In situ repairing done in 2013 by machining of engine block and fitting of assembly fit top liner insert.	No water leakage observed in this engine since block repair work. Approx expenditure was Rs 1.5 crore and time taken for restoration was 125 days. MLPU de-commissioned in SMPL DBL.

In MPPL engines, earlier KBPL, till date water leakage problem and block crack was observed in three engines only i.e. MP3 Mundra, MP1 MPPL Abu Road and MP1 Beawar, out of the nine suspected engines. In these three engines short term solution methodology mentioned above was adopted. Details are placed below:

SN	Engine	Remarks
1	MP3 Mundra	• First time water leakage was observed in August 2015 in Cylinder#8. Extended sleeve fitted. • No water leakage since then.
2	MP1 MPPL Abu Road	• First time water leakage observed in October 2011 in Cylinder#6, 7, & 8. Extended sleeve fitted in all the cylinders. • Again water leakage observed in cylinder No 8 in Nov 2015. • In April 2017 water leakage observed in Cylinder No 3, 4, 7 & 8.
3	MP1 Beawar	• First time water leakage was observed in 2010. Extended sleeve fitted. • No water leakage since then.

All the nine original KBPL engines (Kandla, Aburoad & Beawar) are susceptible to water leakage due to crack, WRPL may ensure that dry penetration test and Magnetic particle test in these engines will be conducted in the coming schedule maintenance.

5. Short term Solution adopted for MP1 MPPL Aburoad in September 2017

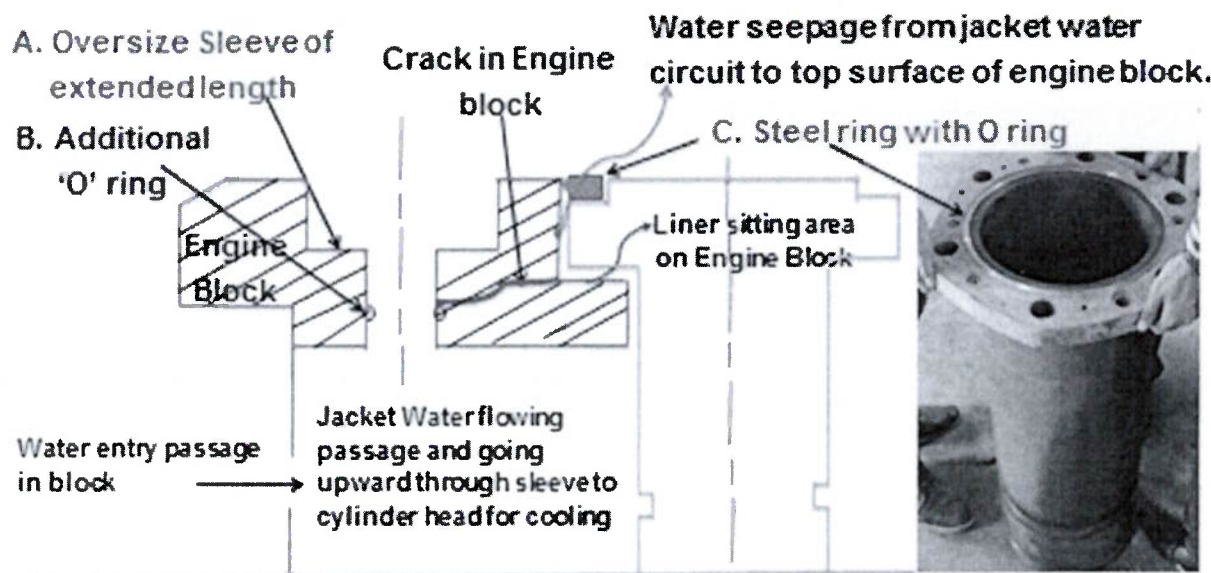
The long term options were time consuming and attending the water leakage problem would have been possible only after 4 to 5 months, a short terms solution was conceptualized. Approximate time taken for restoration would be approximately 8-10 days, expenditure in this case would be Rs 5.0-6.0 lakh and lead time would be approximately 1 month.

The short term solution involves replacement of existing water transfer sleeve with new longer length sleeves and enlarging bores for all the cylinders of block. O-ring is fitted between water transfer sleeve and hole. This will lead to blockage of leakage path around the water transfer holes, hence arresting the leakage of water.

Also sealing the area around the stud at the neck of the hole with a silicone based sealant will help in preventing the corrosion of studs. Since there is also possibility of crack propagation to underneath the top portion of block to water jacket, water seepage to liner landing area through this

path may occur. To prevent this external leakage, modification in steel ring done with Viton O ring fitted in outer dimension of steel ring by making small groove so that water can be stopped from coming out of block.

Schematic Sketch showing proposed modification is placed below.



Repairing work started on 25.08.2017 and hydro static test of engine block done on 06.09.2017 after completion of repairing. Load trail run taken on 08.09.2017 and engine handed over to operation. **Flag-D**

6. Permanent solution of the Problem

- A. **In situ repairing of Engine Block** - Machine 62 mm from the top of the cylinder to remove all the cracks and fit top liner inserts, **similar repair was carried out on Sendra MP-2 in 2013**, repairing time taken was 4 months, apart from 8-10 month lead time for material and service contract.

Approximate expenditure in this case would be Rs 2.0 -2.5 crore along with major overhauling and engine will be out of service for approximately 4 months (minimum).

- B. **Replacement of block with spare block** – Following Four Old G series engines were decommissioned in SMPL DBL project and block of these engines can be reused.
1. MP1 SMPL Sidhpur -
 2. MP2 SMPL Kot
 3. MP1 SMPL Sendra
 4. MP2 SMPL Sendra

MP2 KOT & MP2 Sendra, Old G series engine block can be used as there was no water leakage from block at the time of decommissioning.

However blocks of MP1 Sidhpur & MP1 Sendra cannot be reused without repair as in both engines there was heavy water leakage at the time of decommissioning because of crack in the block.

Details of engine block that can be used are placed below.

SN	Block Details	Remarks
1	SMPL KOT MP2 Block , Old G series, condemned and decommissioned.	• New G series engine block of Kot fire was fitted in 2009 after inspection & testing at CRhrs 73190 hrs with customized water transfer plate supplied by M/s Allen Diesel. • CRhrs of Engine is 99636 hrs. • Operation since replacement of engine block is 26446 hrs. No water leakage, block can be used.
2	SMPL Sendra MP2 Block Old G series engine, condemned and decommissioned.	• In situ repairing done in 2013, at CRhrs 116815, by machining of engine block and fitting of assembly fit top liner insert. • CRhrs of block as on 31.08.2017 is 129457 hrs. • Operation since repair of engine block is 12642 hrs. No water leakage, block can be used.

Approximate expenditure in this case would be Rs 1.0-1.5 crore along with major overhauling of engine and time taken for restoration would be approximately 40 - 45 working days.

- C. **Replacement of block with a new top flat design block** - Lead time for procurement of material and award of service contract would be approximately 12-16 months.

Approximate expenditure in this case would be Rs 4.0-4.5 crore and time taken for restoration would be approximately 40 - 45 working days.

Comparison of all the three options:

SN	Repair Methodology	Time of Non Availability	Total Duration of Execution	Total Cost
1	Option I : In situ repair of engine block	4 months	12 months	Rs 2.0 – 2.5 Cr
2	Option II : Replacement of block with spare block	2 months	4 months	Rs 1.0 -1.2 Cr
3	Option III : Replacement of block with a new top flat design block	2 months	12-16 months	Rs 4.0 – 4.5 Cr

7. Suggestion & Recommendation

In MP1 MPPL Abu Road, water leakage problem was first report in 2011 and subsequently in 2015 & 2017. All the time temporary solution mentioned above was adopted. From the data it can be inferred that the problem is aggregating because the time period between two water leakages decreased.

In view of the above, a long lasting permanent solution of this engine is essential. There are three options available for permanent solution mentioned above.

Option II is more economical and implementation time will be also less. Therefore we propose for replacement of MP1 MPPL Abu Road block with block of decommissioned old G series engine of MP2 SMPL Kot.

Original nine KBPL engines (Kandla, Aburoad & Beawar) are susceptible to water leakage / crack, hence WRPL may ensure that inspection (DP test and MPI) of liner landing area in these engines may be conducted in the forthcoming major maintenance.

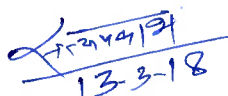
8. In-Principal Approval

In view of the above, In-principal approval may please be accorded for permanent solution of water leakage problem of MP1 MPPL Abu Road. Upon approval, it may please be forwarded to WRPL for further assessment, detailed planning and necessary action.

Submitted for kind perusal please.


(Amit Kumar)
SMNM, PLHO

GM (Maint.)


13-3-18 (S.P. Pandey)

CGM (M&I)

P1. discuss with WRPL and workout a detail plan with timeline along with three documents put up again.

GM (M&I)


19/2/2018

Sub: Cracks found in entablature of MP-1 where liner collar lands

Equipment	MP-1
Location	MPPL, Abu Road
Engine Serial Number	D6/50161-6
Make	M/s. WH ALLEN, UK
Type	8S37G HBC
Running Hrs	51807

Around 15th October water leakage was observed between cylinder head and engine block of cylinder number 6, 7 and 8. Crankcase doors were opened and checked for water leakage inside the liner and since no water leakage was found it was decided to run the engine and take up the issue during 9k maintenance which was only 250 hrs away. No major defect or complaint was observed in the engine since last major maintenance was carried out in Oct-2009.

9k maintenance of the engine was started on 14.11.2011. After removal of heads and liner cracks were found in the entablature where liner collar lands. To ascertain the magnitude and severity of cracks a third party M/s A. R. Inspection Services, Ahmedabad was asked to carry out Dry penetrant test and Magnetic Particle testing.

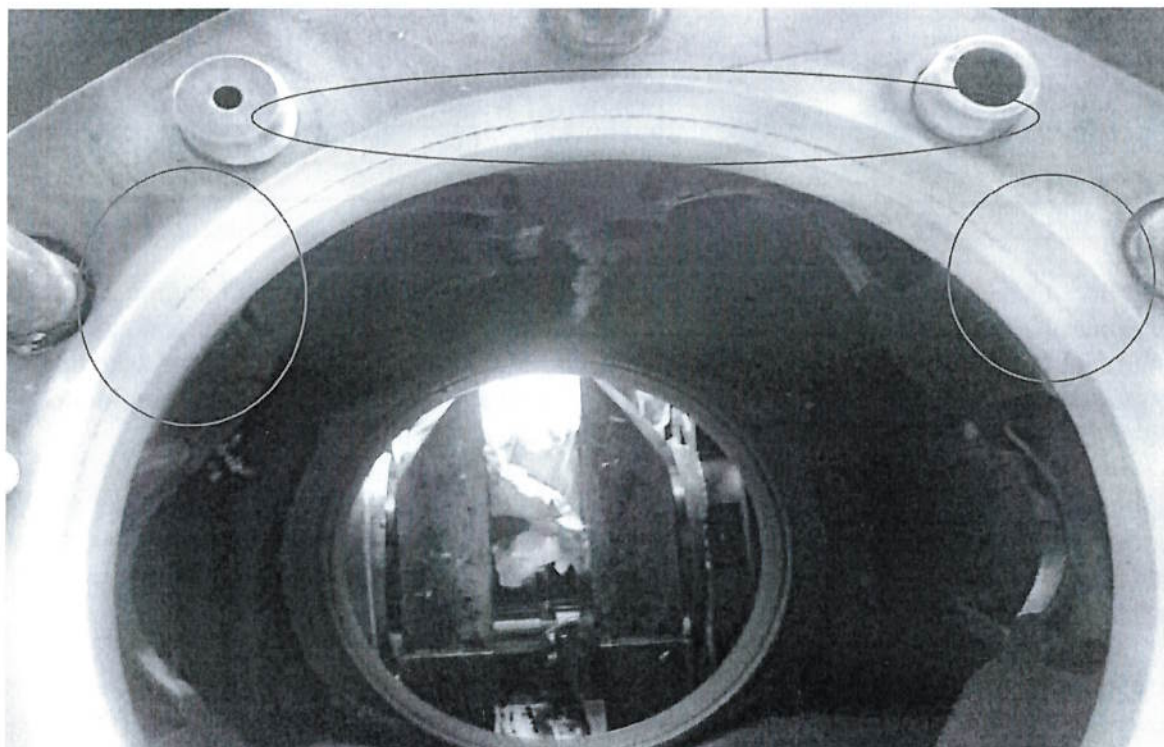
Party carried out the testing on 01.12.2011 and cracks were found along the periphery as shown in figures in all the eight cylinders. Reports are attached as Annexure I.

Technical Solution

As per previous experience of the same problem and consultation held between Abu Road, Viramgam, Central Maintenance-Gauridad, PLHO-Noida and M/s Allen Diesels short term solution of removing old sleeves and inserting new longer length sleeves by enlarging holes will be carried out to reduce downtime of the equipment.

M/s Allwyn industries, Rajkot has been lined up for carrying out the job. The party started the work on 15.12.2011. All old sleeves were removed and new sleeves were fitted by the party on 16.12.2011. Dimensions of the new sleeves are shown in Annexure-II.

Cylinder 1



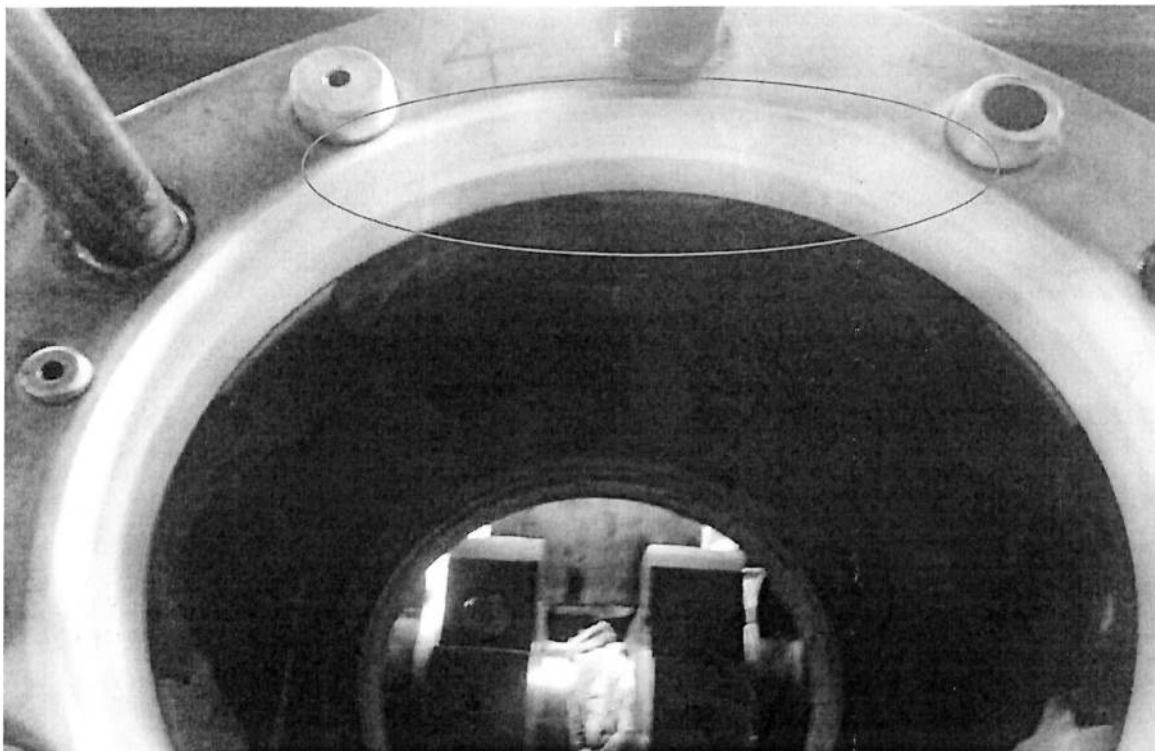
Cylinder 2



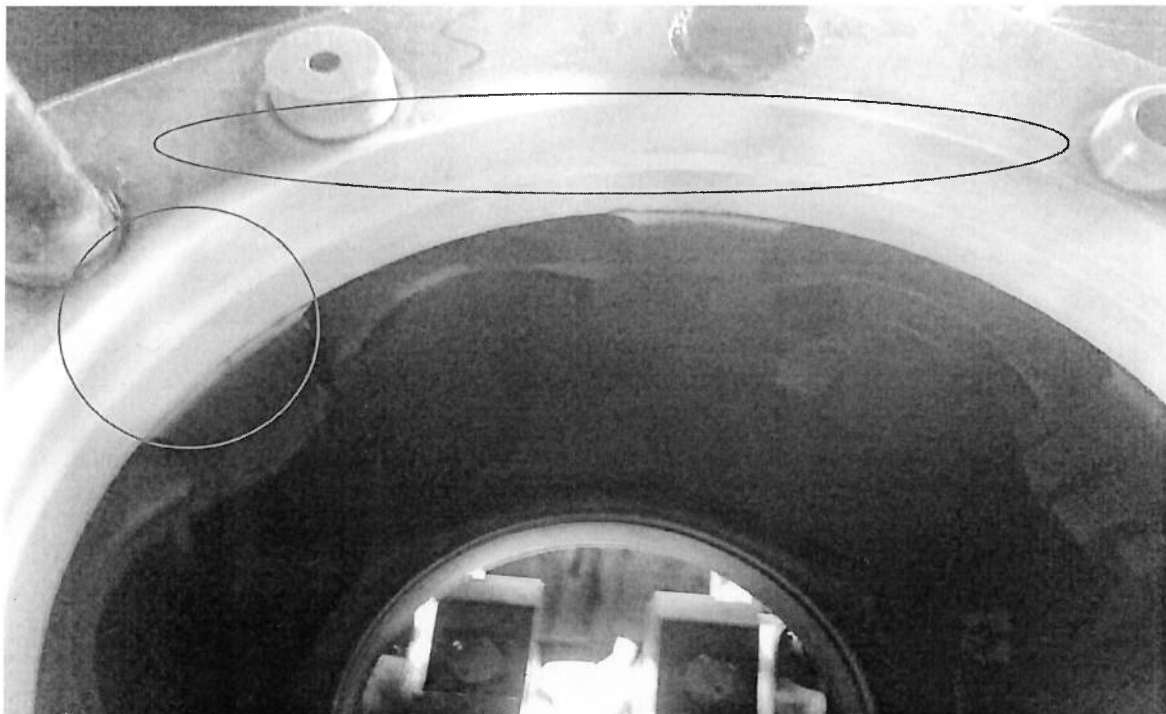
Cylinder 3



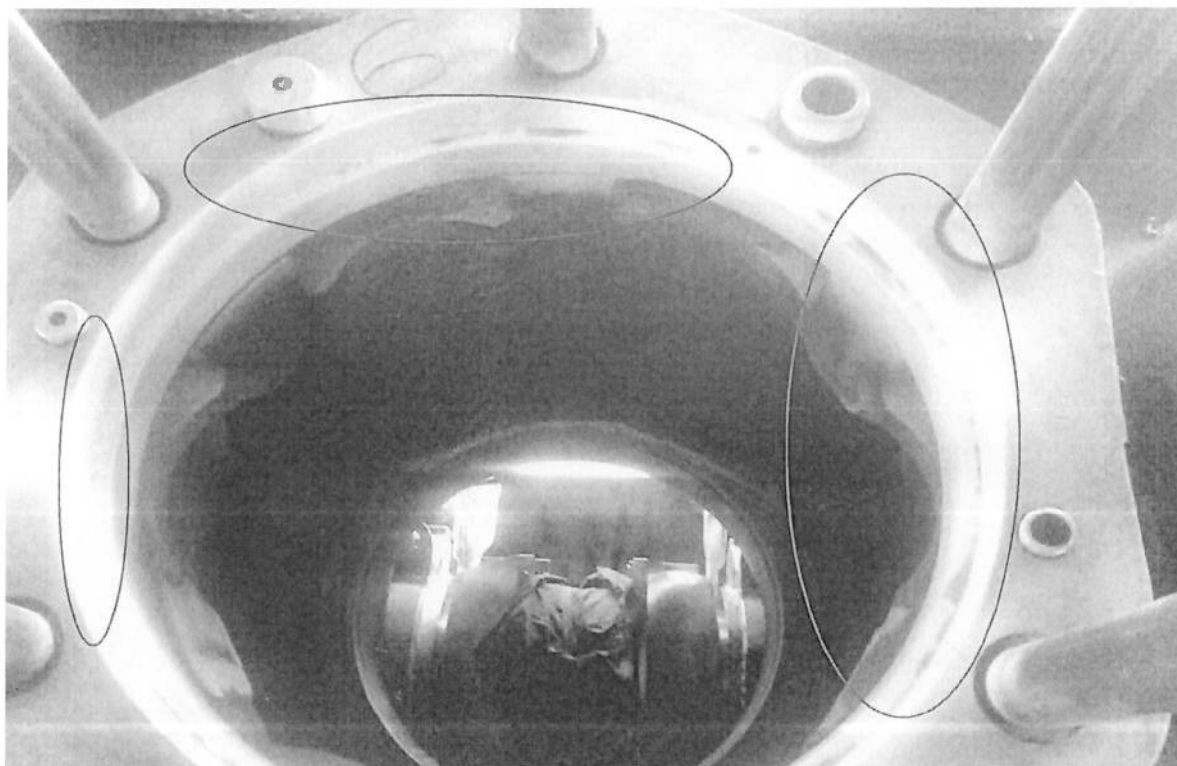
Cylinder 4



Cylinder 5



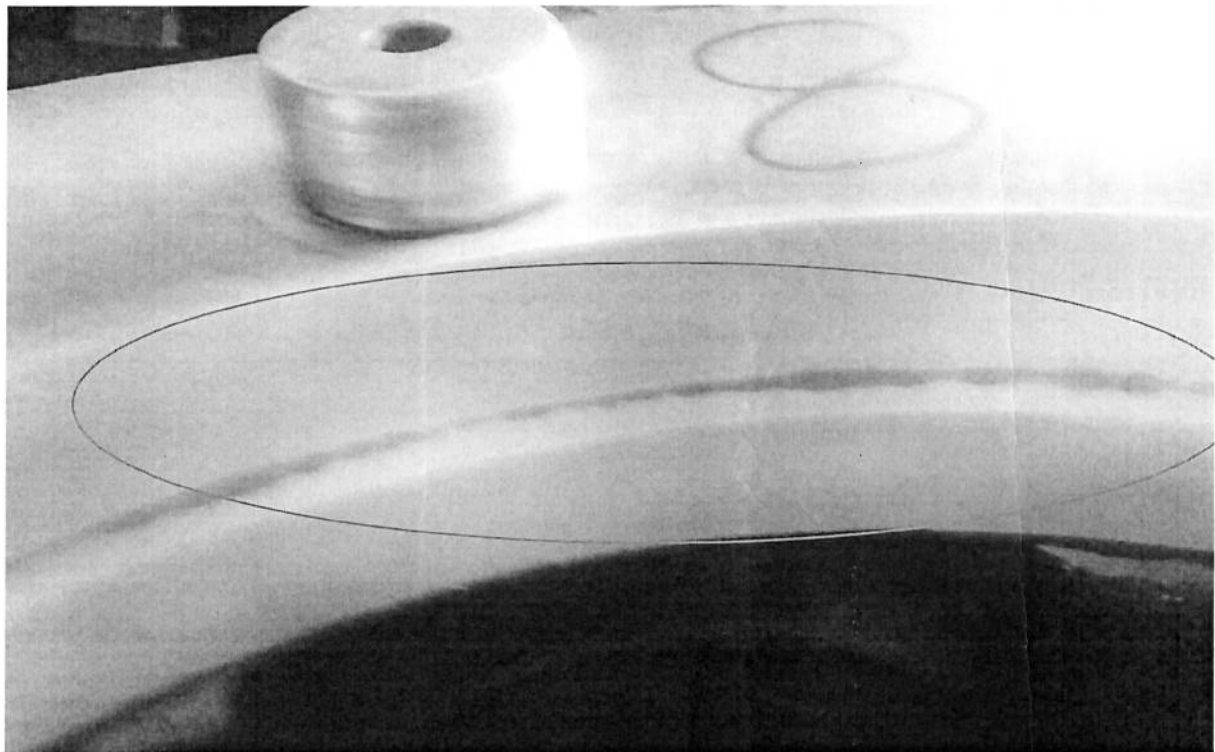
Cylinder 6



Cylinder 7



Cylinder 8



Liquid Penetrant Testing Report



A.R. INSPECTION SERVICES

59, Nilakanth Avenue, Nr. Kailash Estate, Nikol-Odhav SP. Ring Road, Ahmedabad-382 415 (INDIA)
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REPORT NO: PT/IOC-111202

DATE: 01/12/2011

LIQUID PENETRANT TESTING REPORT

CLIENT : M/S INDIAN OIL CORPORATION LTD.
SITE : WESTERN REGION PIPELINES, ABU ROAD
JOB : EQUIPMENT : MP-1 LOCATION : MPPL, ABU ROAD
SR. NO.: D6/50161-6 TYPE: 8S37G HBC
REF.STANDARD : (A) TECH.SPECIFICATIONS OF IOCL SEC.1 FOR L.P.T.
(B) ASTM E-165 "RECOMMENDED PRACTICE FOR L.P.T."
(C) ASME SEC V ARTICLE-6"

TESTING PARAMETERS:

Material used	Cleaner	Penetrant	Developer
	Make PMC PC-120 B BT. NO. PT 23 20325 EXP. DT. JAN 2014	Make PMC PP-110B BT. NO. PT 10 9245 EXP. DT. OCT 2013	Make PMC PD-130B BT. NO. PT 23 62830 EXP. DT. JAN.2014
Test Temp.	ABOVE 15°	White Light Intensity	Min. 100 fc
Method used	Visible solvent removable		
Dwell time	10 Min.	Examination time	With in 7 min. after applying developer

SCOPE OF TEST.: TESTING ACCESSIBLE AREA ENTABLATURE BODY WHERE LINER COLLAR SITS

OBSERVATION : THERE WERE CIRCUMFERENCIALY DISCONTINUITIES INDICATIONS OBSERVED FROM TESTING ACCESSIBLE AREA OF PARTS (08 NOS)

REMARKS : ABOVE MENTIONED "PARTS (08 Nos)" WERE FOUND UNSATISFACTORY IN LIQUID PENITRANT TESTING.

TESTED BY : SHIVKUMASR G. TOMAR (ASNT LEVEL-II PT,MT)
DATE OF TESTING: 1ST, DEC., 2011

FOR, A.R.INSPECTION SERVICES

A. Thakur
J.A. THAKUR
A.S.N.T.LEVEL-II UT,RT,MT,PT



WITNESSED BY
Mr. AASHISH GUPTA (O & ME)
W.R.P.L., ABU ROAD I.O.C.L.

Magnetic Particle Testing Report



A.R. INSPECTION SERVICES

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Mobile : 98240 61508, 99246 49071 • E-mail : arinspection@yahoo.com

REPORT NO:MT/IOC-111202

DATE: 01/12/2011

MAGNETIC PARTICLE TESTING REPORT

CLIENT : M/S INDIAN OIL CORPORATION LTD.
SITE : WESTERN REGION PIPELINES, ABU ROAD

JOB : EQUIPMENT : MP-1 LOCATION : MPPL, ABU ROAD
SR. NO.: D6/50161-6 TYPE: 8S37G HBC

PARAMETERS : (A) EQUIPMENT TYPE :ELECTRO MAGNETIC YOKE MAKE: SIMS Y A/D
(B) TYPE OF MAGNETIZING YOKE : 'C' TYPE ADJUSTABLE
(C) MAGNETIZING CURRENT : HWDC
(D) LIFTING CAPACITY :18.5 Kg. IN D.C MODE
(E) MAGNETIC POWDER : WET VISIBLE (SIMS)
(F) VERIFICATION OF FIELD STRENGTH :BY USING PIE GUAGE FIELD INDICATOR

REF.STANDARD: (A) TECH SPECIFICATIONS OF IOCL SEC.2 FOR M.P.T.
(B) ASTM E-709 "RECOMMENDED PRACTICE FOR M.P.T."
(C) ARTICLE-7 "M.P.T. ASME BOILER & PRESSURE VESSEL CODE SEC.V

PROCEDURE : EACH TESTING ACCESSIBLE PORTION OF PART WAS SPRAYING WHITE CONTRAST AND MAGNETIZING SPACING BETWEEN TWO LEGS OF YOKE TO BE ADJUSTED TO SUIT DIFFERENT LOCATION BUT MAX. SPACING FOR ANY LOCATION WAS 150 mm. MAGNETIZING AREA AND FLOWING WET VISIBLE PARTICLE WAS SYNCHRONIZED FOR EFFECTIVENESS OF DISCONTINUITIES DETECTION.

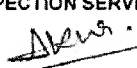
SCOPE OF TEST.: TESTING ACCESSIBLE AREA ENTABLATURE BODY WHERE LINER COLLAR SITS

OBSERVATION : THERE WERE CIRCUMFERENCIALY DISCONTINUITIES INDICATIONS OBSERVED FROM TESTING ACCESSIBLE AREA OF PARTS (08 NOS.)

REMARKS : ABOVE MENTIONED "PARTS (08 Nos.)" WERE FOUND UNSATISFACTORY IN MAGNETIC PARTICLE TESTING.

TESTED BY : SHIVKUMASR G. TOMAR (ASNT LEVEL-II PT,MT)
DATE OF TESTING: 1ST,DEC.,2011

FOR, A.R.INSPECTION SERVICES


J.A. THAKUR
A.S.N.T.LEVEL-II UT,RT,MT,PT



WITNESSED BY
Mr. AASHISH GUPTA (O & ME)
W.R.P.L., ABU ROAD I.O.C.L.

**INDIAN OIL CORPORATION LIMITED
WESTERN REGION PIPELINES
BREAKDOWN ANALYSIS REPORT**

EQUIPMENT: MP-1

LOCATION: MPPL, Abu Road

RUNNING HRS. 67290:05

Date: 15.12.2015

Sub: Analysis report of external water leakage from cyl no 8 of MP-1 ,MPPL at WRPL-Abu Road

OBSERVATIONS DURING BREAKDOWN:

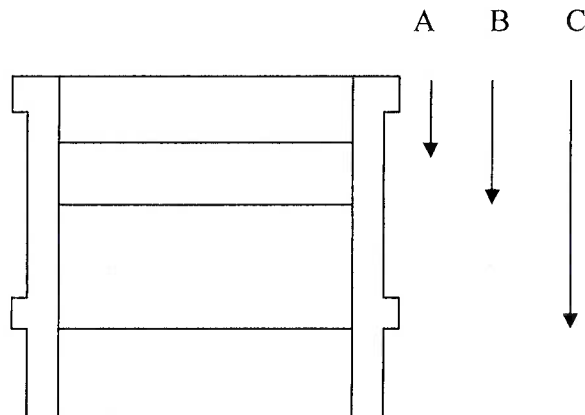
1. External water leakage was observed between cylinder head and entablature of cylinder no-8 on 24.11.2015. The 6000 HRS schedule maintenance was going to commence soon so engine was taken in schedule maintenance.

PROPABLE REASONS FOR BREAKDOWN:

1. It was suspected that water leakage could be due to following reasons:
 - a) During the 9000HRS maintenance of the MP-1 in year 2011, cracks were found in the entablature (in all cylinder blocks) where liner collar landing area. At that time after suggestion from M/s Allen Diesels, short term solution of removing old sleeves and inserting new longer length sleeves by enlarging holes was done. Now we are suspecting that either the sleeves are not holding the water or the old cracks in block are propagated further resulting in water leakage.
 - b) The contact area between the liner and block may have decreased and water may have leaked from mating area.
 - c) The clearance between cylinder head and entablature is not sufficient for proper tightening of cyl head.

MAITENANCE STEPS CARRIED OUT:

1. Cylinder head assembly of cyl no-8 engine was lowered and no water content was found on piston and liner area. Water found on cylinder block area.
2. Cylinder head was hydro tested and found ok. It was suspected that water leakage was from water transfer O rings so all the o rings were replaced. Same cylinder head was fitted again and jacked water circuit was pressurised. Again the external water leakage was persisting.
3. The new cyl head & steel ring was installed and clearance between cyl head & entablature was coming around 2.5mm which was satisfactory. The clearance was compared with other cyl heads also and found ok. But the external water leakage was still persisting.
4. Now it was decided to take out the liner for block inspection. Liner , piston and connecting rod were taken out. Blue contact matching was done for the block and contact area was found around 25%. lapping of about 0.1mm was carried out using lapping tool and contact area was increased around 90%.
5. New liner was received from Viramgam. The old liner was checked for dimensions and ovality and found ok. Dimensions are mentioned below:



Point-A is 100 mm from TOP of liner

Point-B is 400 mm from TOP of liner

Point-C is 600 mm from TOP of liner

LINER MEASUREMENTS

Cylinder no	Direction	Cyl liner No-8
Position A	X	325.16
	Y	325.18
Position B	X	325.15
	Y	325.17
Position C	X	325.16
	Y	325.16
Max Ovality		0.02

All dimensions are in mm

- All the piston ring were replaced. All the side clearances for pistons were taken found in limits. Side clearance in grooves of top pr ring, 2nd pr ring, 3rd pr ring and scrapper ring was 0.23, 0.13, 0.13 and 0.05 mm respectively (max permissible clearance is 0.3mm).
- The water was coming from one side of cylinder head area so suspected sleeve was taken out and refitted by applying loctite compound.
- Now the old liner with new liner o rings, piston and connecting rod assembly, cyl head assembly were fitted back. Jacket water circuit was pressurised and external water leakage is persisting till date.
- The jacket water network was charged for 2-3 hours and water leakage inside the engine was checked and no water content found (checked from injector sleeve).
- Assistance from base/region was sought for permanent solution of this problem. Replacement of all water transfer sleeves of cyl no.8 was planned and M/s Manan Engineering Corporation, Rajkot was lined up for this work. Party manufactured the sleeve and installed in cyl no.8 of engine.
- After replacement of sleeves, engine trial run was taken and no water leakage was observed. The engine was taken in observation for 24 hours and no water leakage observed. Now the engine is

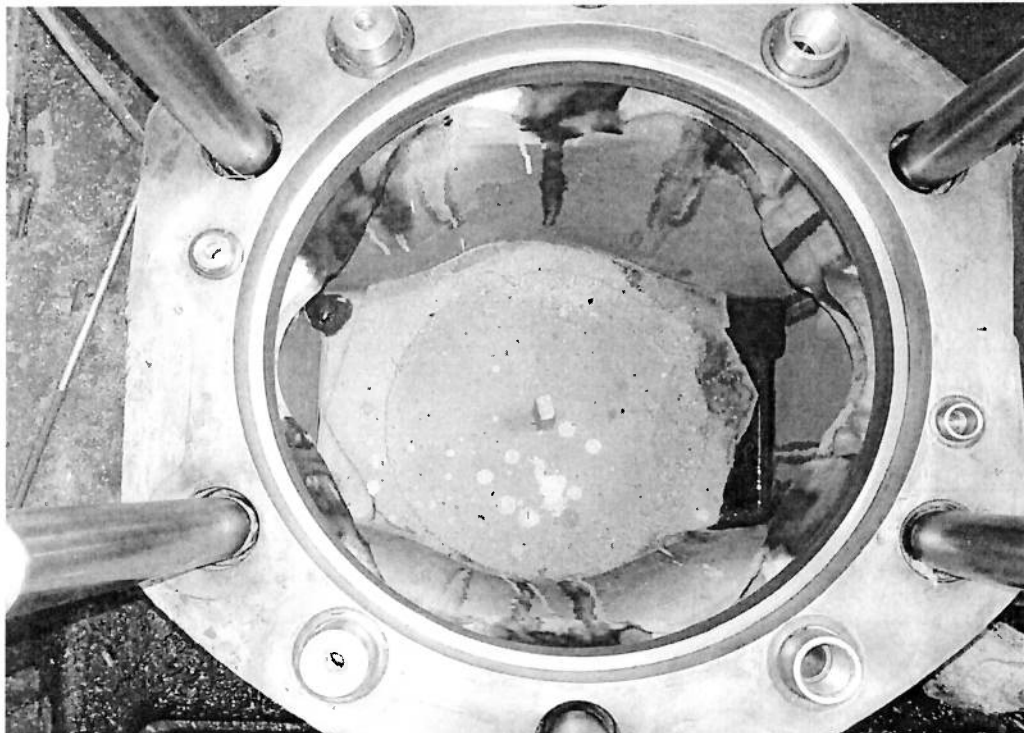
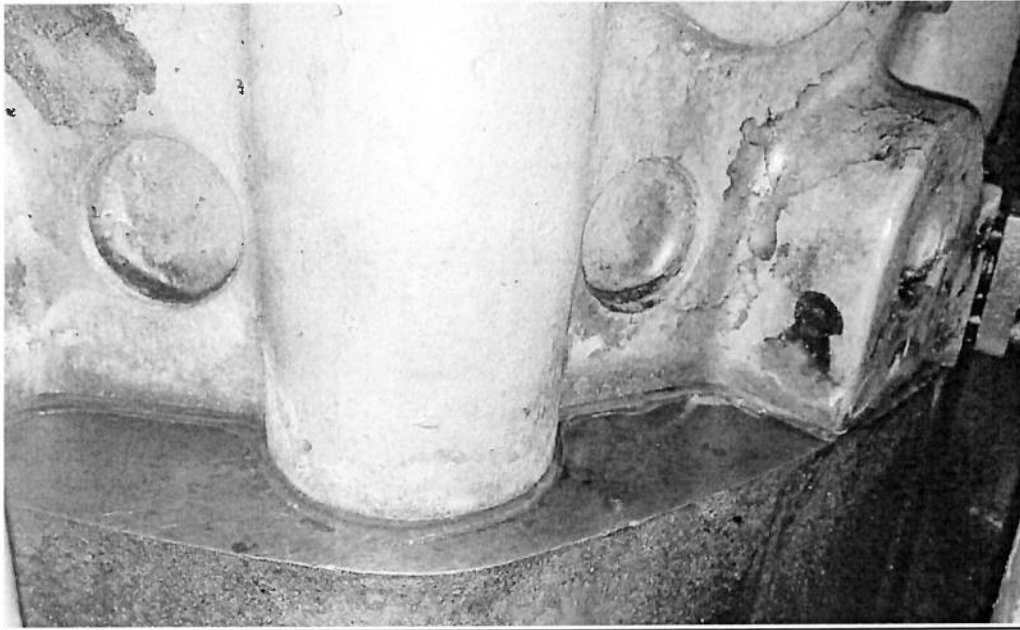
running satisfactorily.

Enclosed:

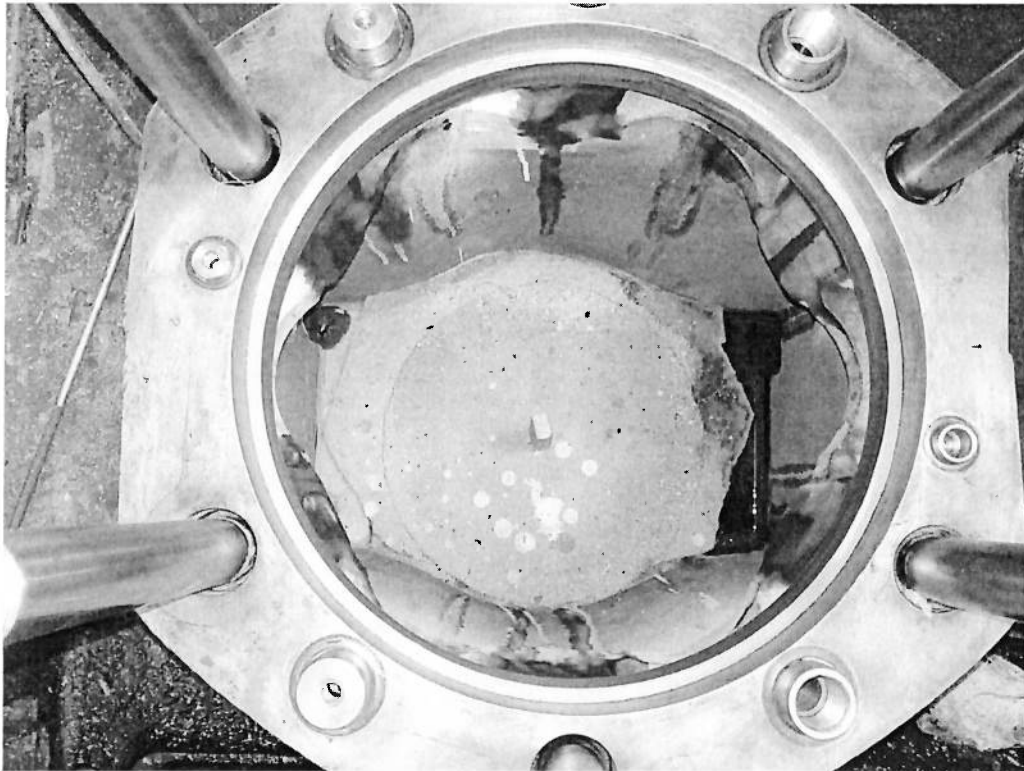
1. Photographs during maintenance are enclosed below
2. Analysis report for sleeving job carried out during 9000HRS major maintenance in 2011

Photographs:

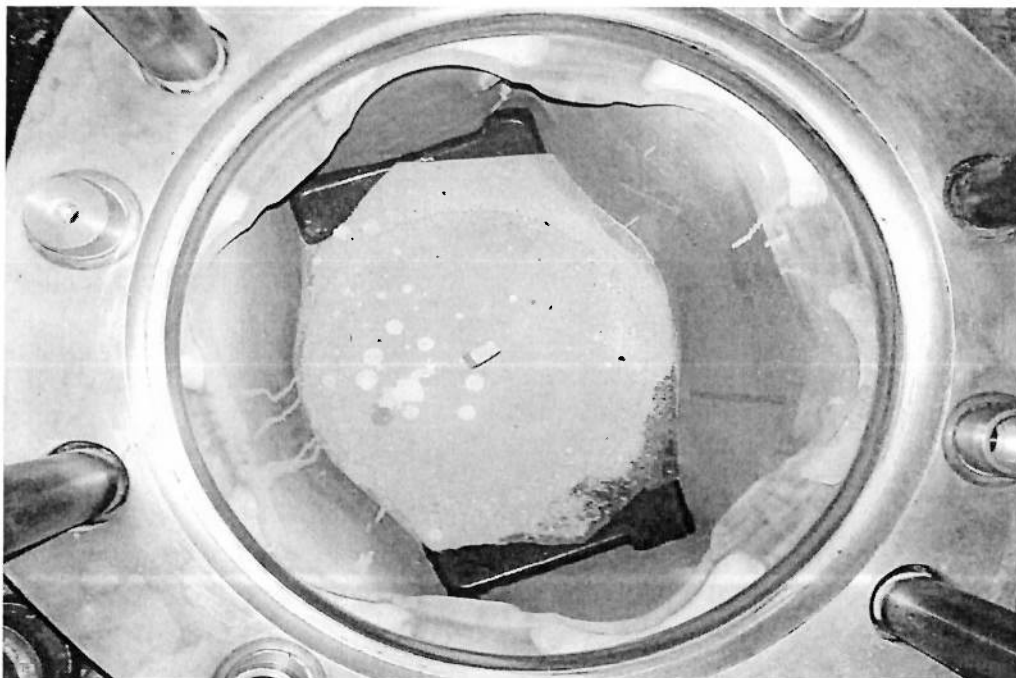
1. Initial water leakage observed in cylinder block



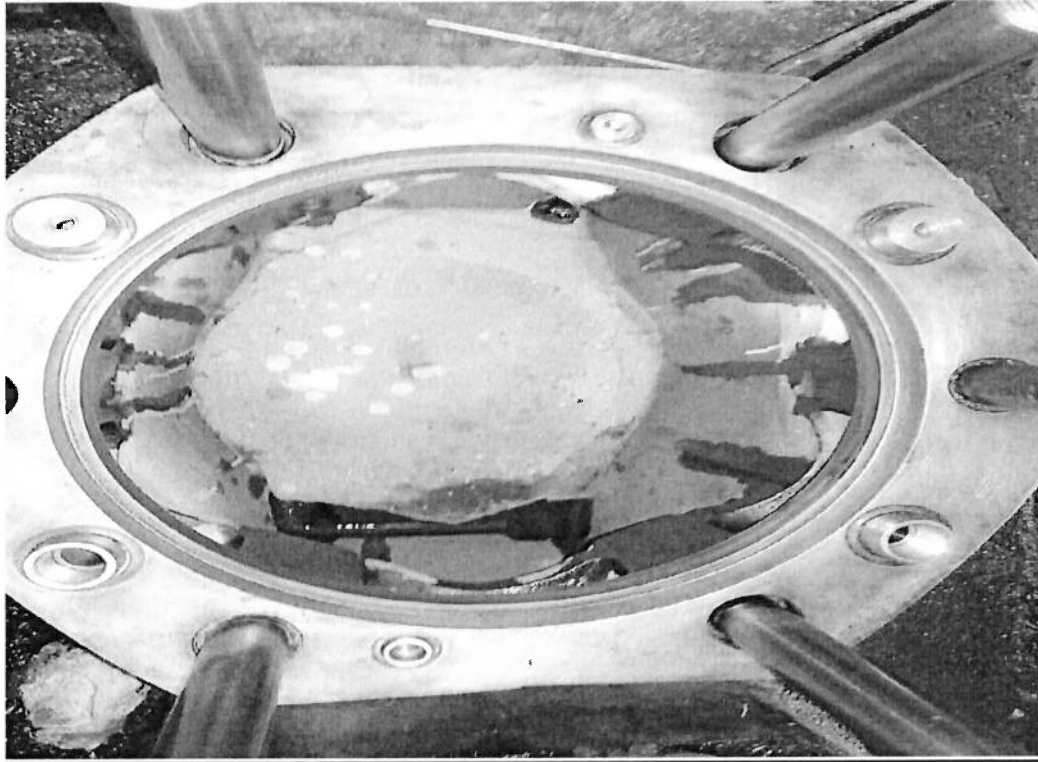
2. Cracks observed in cylinder block



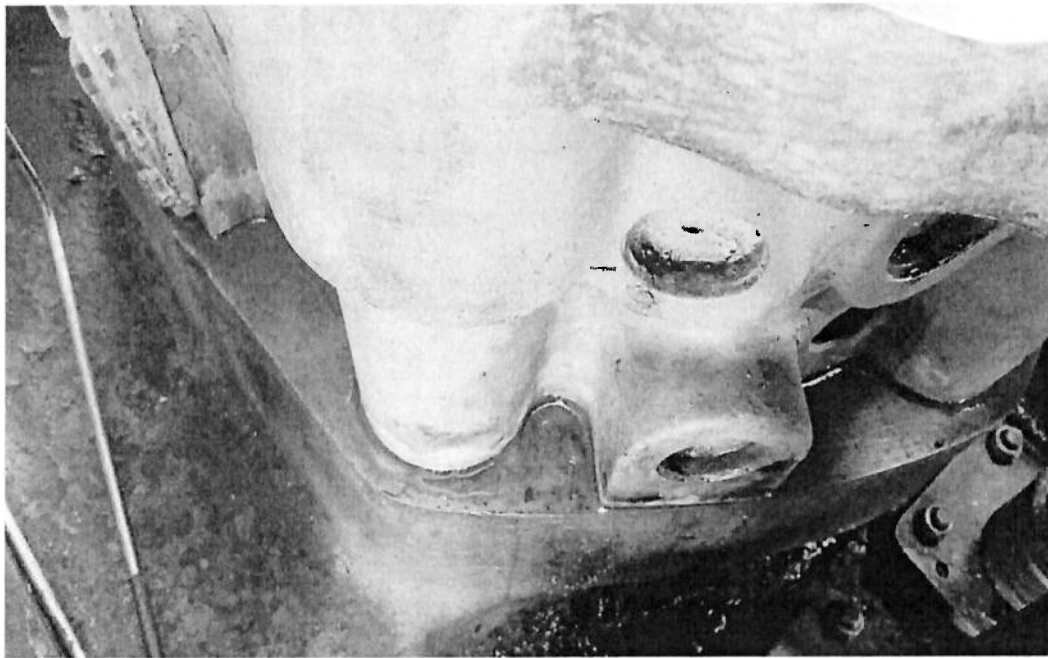
3. Contact area by blue matching of block in liner landing area



4. Blue matching after lapping of block area



5. Water leakage observed during final inspection on 08.12.2015





Allen Incident Report	Date: 11 th April 03 Report No. 4000 IOCL 2003 / 001
Client	IOCL
Site and Set No.	Sidhpur MP-2
Engine Type	Allen S37G
Date of Incident	10 th April 2003
Engine Serial No.	D5/62982/B
Operating Hours	51,359 Hrs
Reason For Investigation	Water Leakage from Top Liner Landing, Cylinder No. 1

Description of problem and Investigation

IOCL reported leakage of water from between cylinder head No. 1 and the engine block which could not be arrested by changing water transfer tube 'o' rings.

Mr Ashish Puri visited site with IOCL personnel 10th April 2003 and has reported as follows:

Cracking has been found on the cylinder block as shown on the attached Figure 1. There are two peripheral cracks in the corner of the locating area in the block in contact with the liner top flange. There are two cracks of around 100mm and 75mm in length detected by plugging the water transfer pipes and introducing compressed air into the water transfer passage from the pocket inside. In addition there are vertical cracks from five of the six cylinder head stud holes to the liner landing bore.

The photograph in Figure 2 also illustrates the location of the cracks.

It was noted that JW pressure on this engine was lower than other engines at site, 1.3 to 1.5 bar since last one year as compared to 2.1-2.4 bar MP1 and 1.7- 1.9 bar MP-3. Log sheets indicate that the engine commonly operates with JW outlet temperatures of up to 85/86 deg C.

As a precaution cylinder nos. 4 & 8 on this engine were checked for peripheral cracks by plugging the water transfer pipes and introducing compressed air into the water transfer passage from the pocket inside. Dye penetrant testing of the liner land area was also carried out. No leaks were detected on these cylinders.

Technical Discussion

The cracking of the cylinder block has most likely been caused by high hoop stresses in the liner seating area caused by an earlier incident of overload or overheating. At this time the cracks would have been instigated and it may have been a long period of propagation until the peripheral cracks reached the water transfer holes. On reaching the water transfer holes a leakage path for Jacket Water is provided by these cracks.

A service bulletin Number 016/4000 was issued advising that liner flange dimensions should be checked as early liners were a bit tighter in the block which we found was more susceptible to cracking of the block in cases of overload or overheating. The liner design has been changed to provide more clearance in the liner landing and the new dimensions are as marked up on the attached Figure 3.

Technical Solution

The cracks observed are likely to continue to propagate increasing the amount of water leakage, however it is not expected that either of these two types of crack would lead to any catastrophic failure of the engine if the set continues to operate. From this respect it is possible to continue operating the engine in its present condition. The practical problem will be the leakage of jacket water which will have to be somehow collected and the loss of water has to be topped up to avoid water starvation. This also presents problems of maintaining correct dosage of treatment chemicals.

Temporary Repair

A temporary repair can be carried out to reduce or stop water leakage. This would involve three steps:

1. Use heat to evaporate all water from all of the peripheral and vertical cracks and coat with Loctite 290 penetrating sealant in accordance with the manufacturer's instructions. It is important to treat the vertical cracks as although they are not a source of water leakage now, they may become a problem later if they propagate past the liner flange seating area.
2. Using a drill and reamer open out the affected water transfer holes and fit a sleeve with Loctite 270 to block the leakage passage from the water transfer hole to the peripheral crack. There are two sizes of water transfer holes as shown in figure 4, water transfer holes adjacent to the peripheral cracks should be opened out from 12 to 16mm or 19.1 to 23mm as appropriate. Sleeves can be manufactured from stainless steel as shown in Figure 5 and 6. These should be a sliding fit in the increased diameter water transfer holes. These should be coated liberally with Loctite 270 and fitted. The water transfer ferrules can be replaced and the clearance between ferrule and cylinder head checked. It

may be necessary to file 2mm off of the top of the ferrule to avoid contact in the water transfer recesses in the head.

3. To prevent water collecting in cylinder head stud holes – which could later lead to corrosion and cylinder head stud failure – seal the area around the stud at the neck of the hole with a silicone based sealant.

It is important prior to replacing the liner it is checked to ensure that the top flange dimensions are in line with the values given in figure 3. In case they are larger then the liner should be machined to within this tolerance. Any new liner supplied from around 1994 onwards should be within this tolerance, so it may be faster to fit a new liner.

Long Term Repair

Since the cracks are likely to continue to propagate the above temporary repair may not last too long before the cracks have created another leakage passage. A longer term repair to rectify this fault has been successfully carried out on another installation where a 60 mm approx. section at the top of the affected cylinder turret is machined off in situ – see Figure 7. A pre-machined insert with an 'o' ring on the bottom surface is fixed to the turret with cap screws bringing the liner landing back to nominal dimensions. R-R can arrange for this repair to be carried out if required.

New Cylinder Block

A further alternative would be to provide a new cylinder block. Note that a new block would be provided to the latest design as supplied for SMPL augmentation and later. This incorporates a flat seating area for the top liner flange and a transfer plate between cylinder head and block. The cylinder heads fitted to SID-MP-2 are not identical to those fitted to this later design cylinder block however a modified transfer plate is available to allow fitting of an older design cylinder head to the new design block. Please note that in case of fitting a new cylinder block to an existing bedplate there is normally a need to shim or use additional jointing material to seal the free end frame and the crankshaft end seal plate. This is due to the bedplate and cylinders being machined as a matched set during manufacture making it difficult to exactly match two individually machined components.

Recommended Actions

This Engine:

Carry out the temporary repair as described above.

Ensure cylinder liner flange dimensions are in accordance with Figure 3.

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Ford End Road
Bedford MK40 4JB

Try to ensure that Jacket Water temperature at outlet from the engine is maintained at the design 82 deg C and check that this temperature is achieved at all individual cylinder head stand pipes.

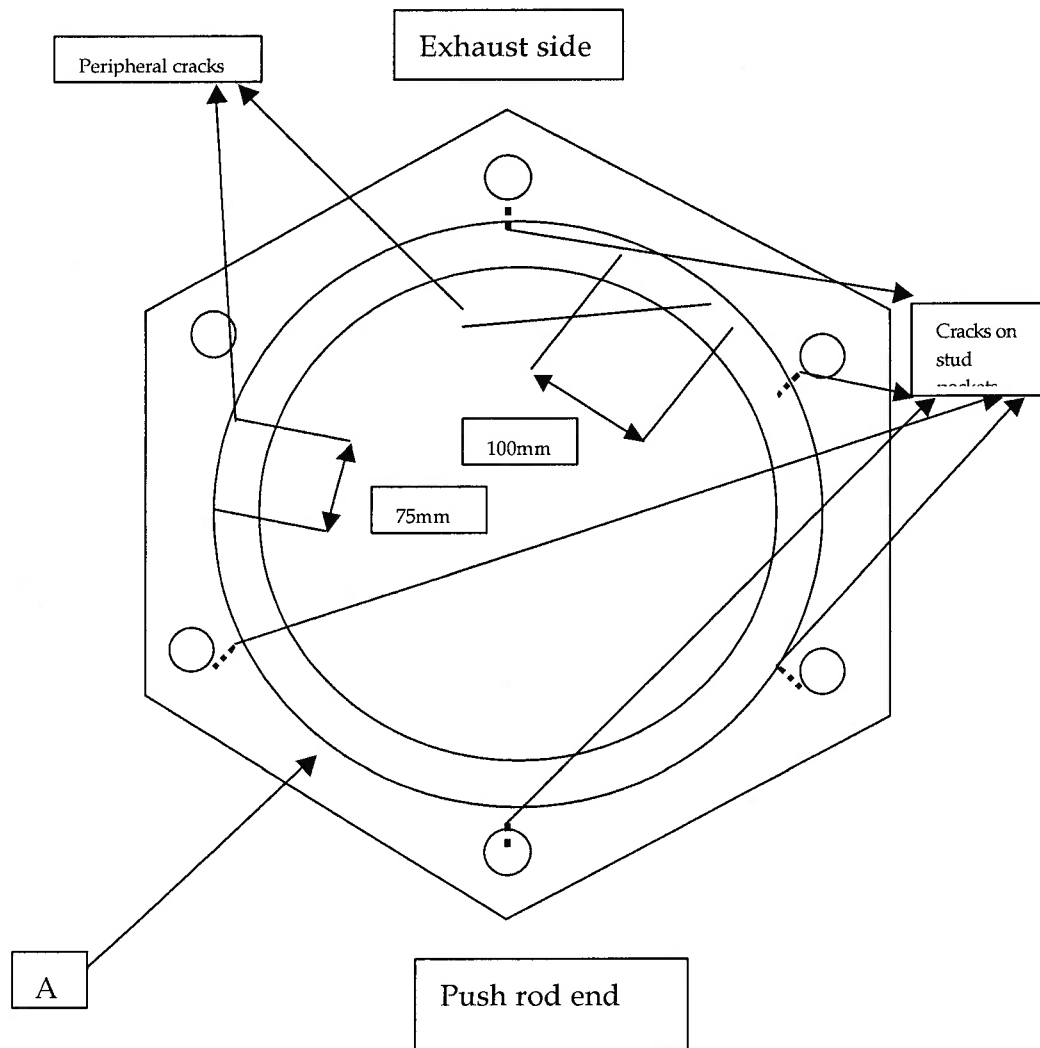
Decide whether a Long Term Repair as above will be carried out or the cylinder block will be replaced and make necessary arrangements while the engine continues to operate with the temporary repair.

Other Engines:

Ensure that the liners on the other "old G series" engines supplied in 1987 for Sidhpur, Kot and Sendra and the original 9 KBPL engines at Kandla, Abu Road and Beawar have top flange dimensions in line with Figure 3.

B J Kane
11/04/03

Figure 1



Note: The cracks on the stud pocket could be detected only after DPT. The cracks exist on the shortest line from the stud hole to the circle A. The cracks are visible on the inner surface of this diameter of circle A but do not reach down the the inner land flange (2/2 ed)

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Queens Engineering Works
Ford End Road
Bedford MK40 4JB

Figure 2 – Photograph of cracked liner landing

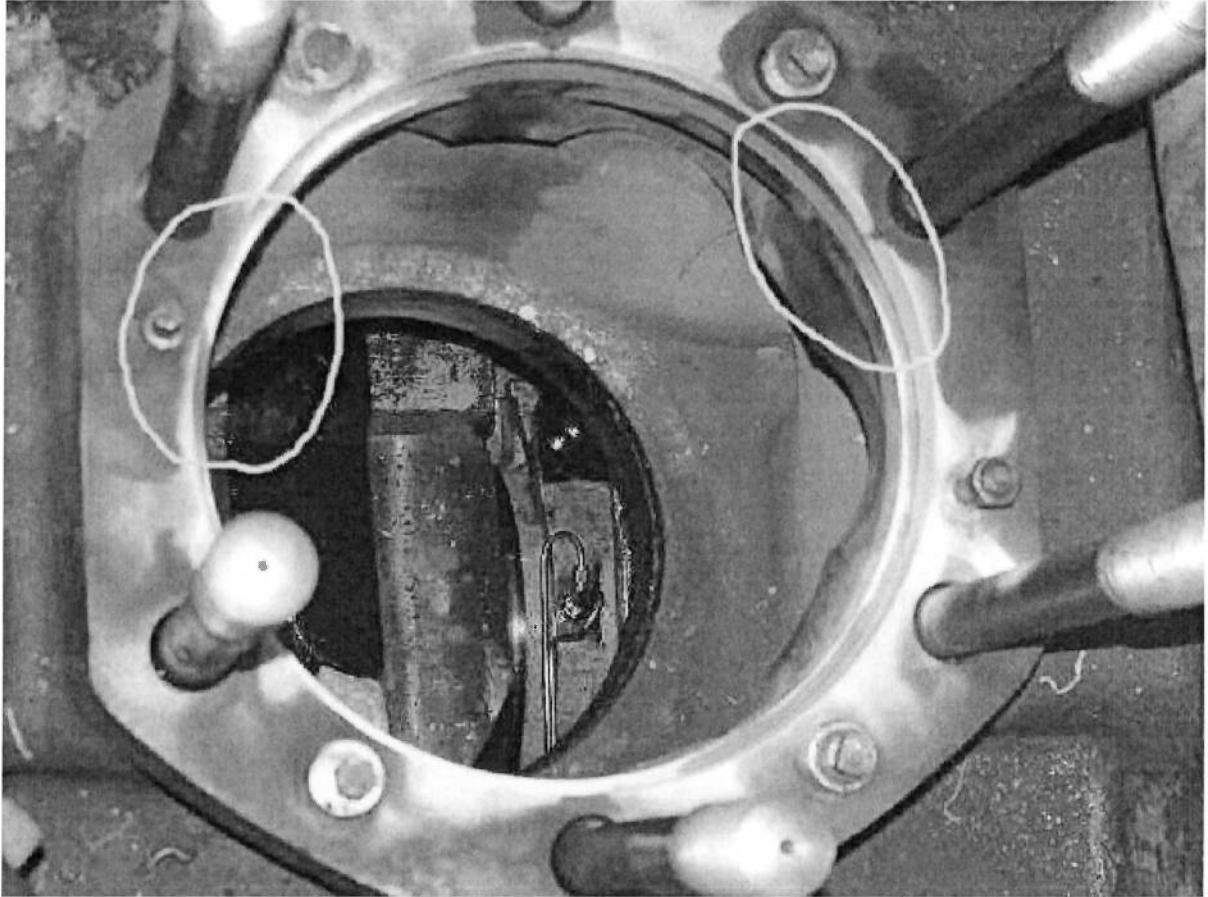
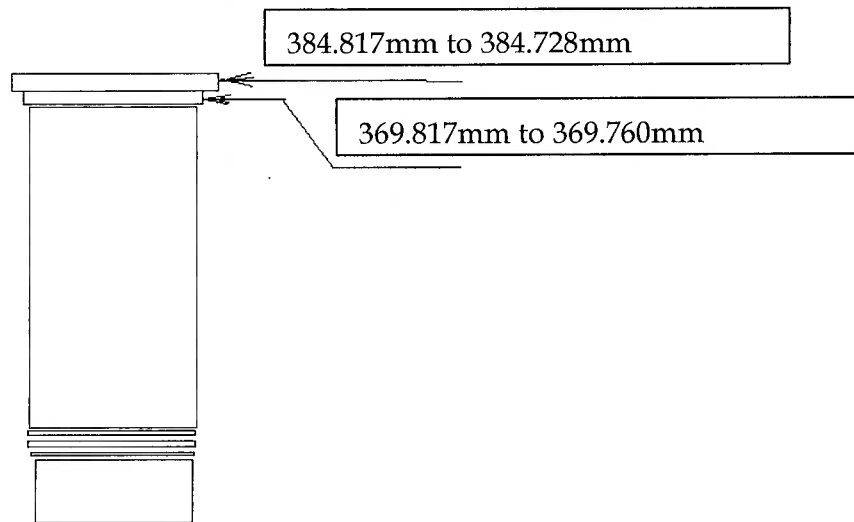


Figure 3

Revised Liner Dimensions



Note that RRPE records show that liners of these dimensions would have been supplied to SMPL Augmentation and later sets. The original six "old G series" engines for SMPL and the 9 original KBPL engines would have been supplied with older liners which had slightly larger flange diameters than above – these should be checked and if necessary machined to within the above tolerances.

Figure 4 Dimensions of Water Transfer Holes

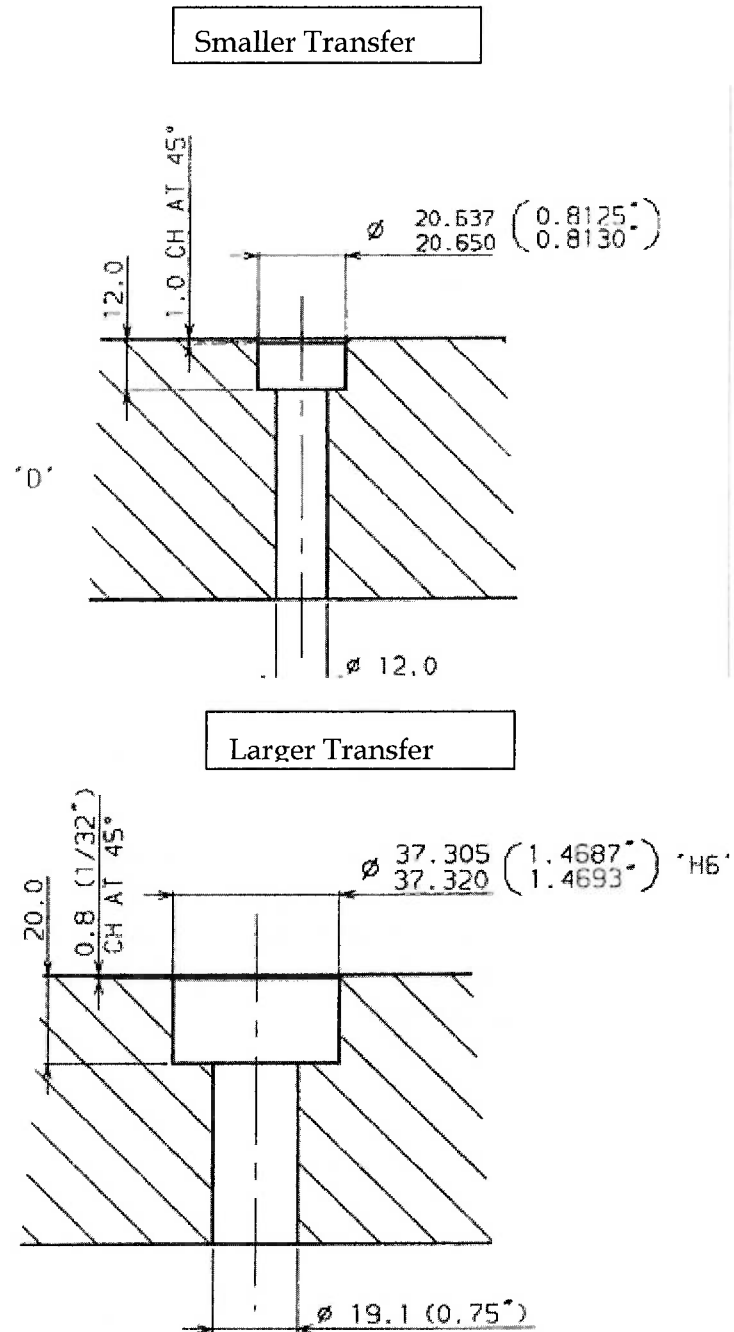
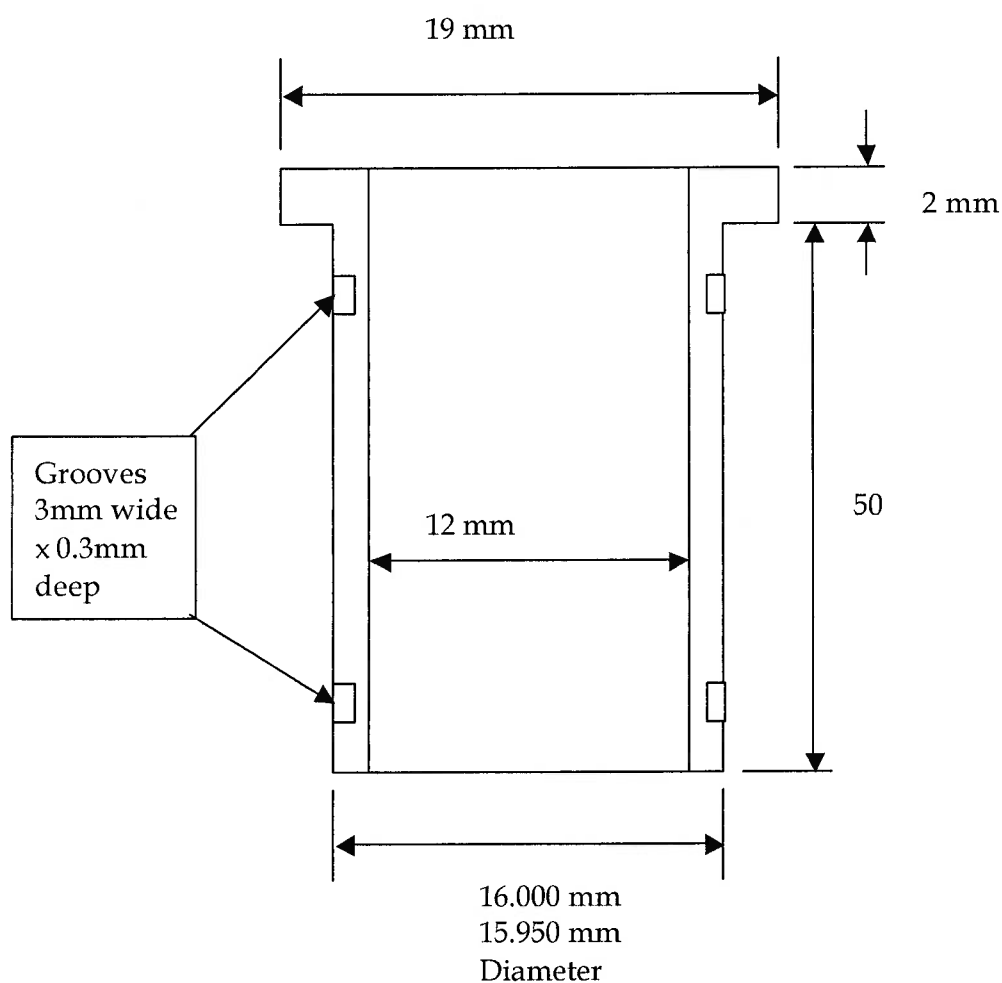


Figure 5 Temporary Repair - Sketch of proposed sleeve to be fitted to smaller water transfer hole.

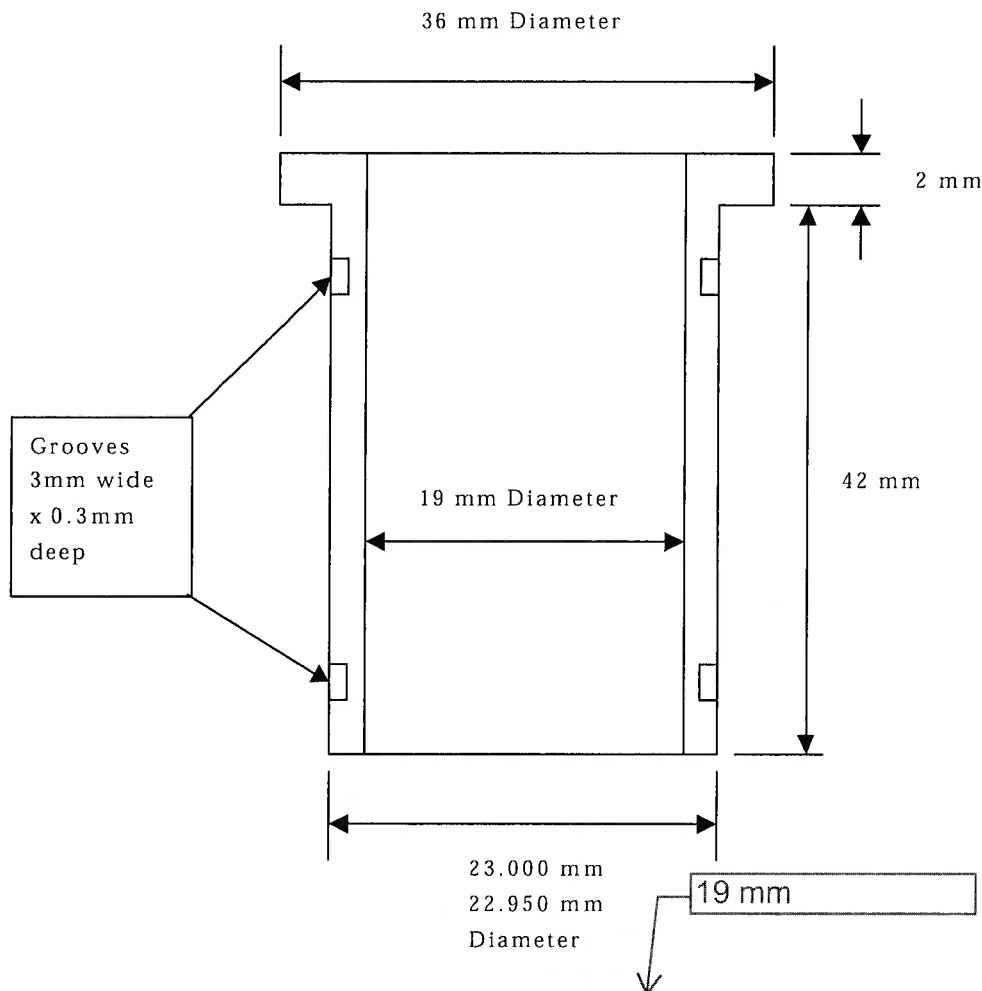


Open water transfer hole from 12 mm diameter to 16.000 to 16.025mm diameter. (aiming for a 0.025mm clearance of sleeve in hole).

Manufacture sleeve from stainless steel as above with grooves near top and bottom 3mm wide x 0.3mm deep for Loctite to collect.

Fit sleeve to transfer hole using loctite 270.

Figure 6 Temporary Repair - Sketch of proposed sleeve to be fitted to larger water transfer hole.

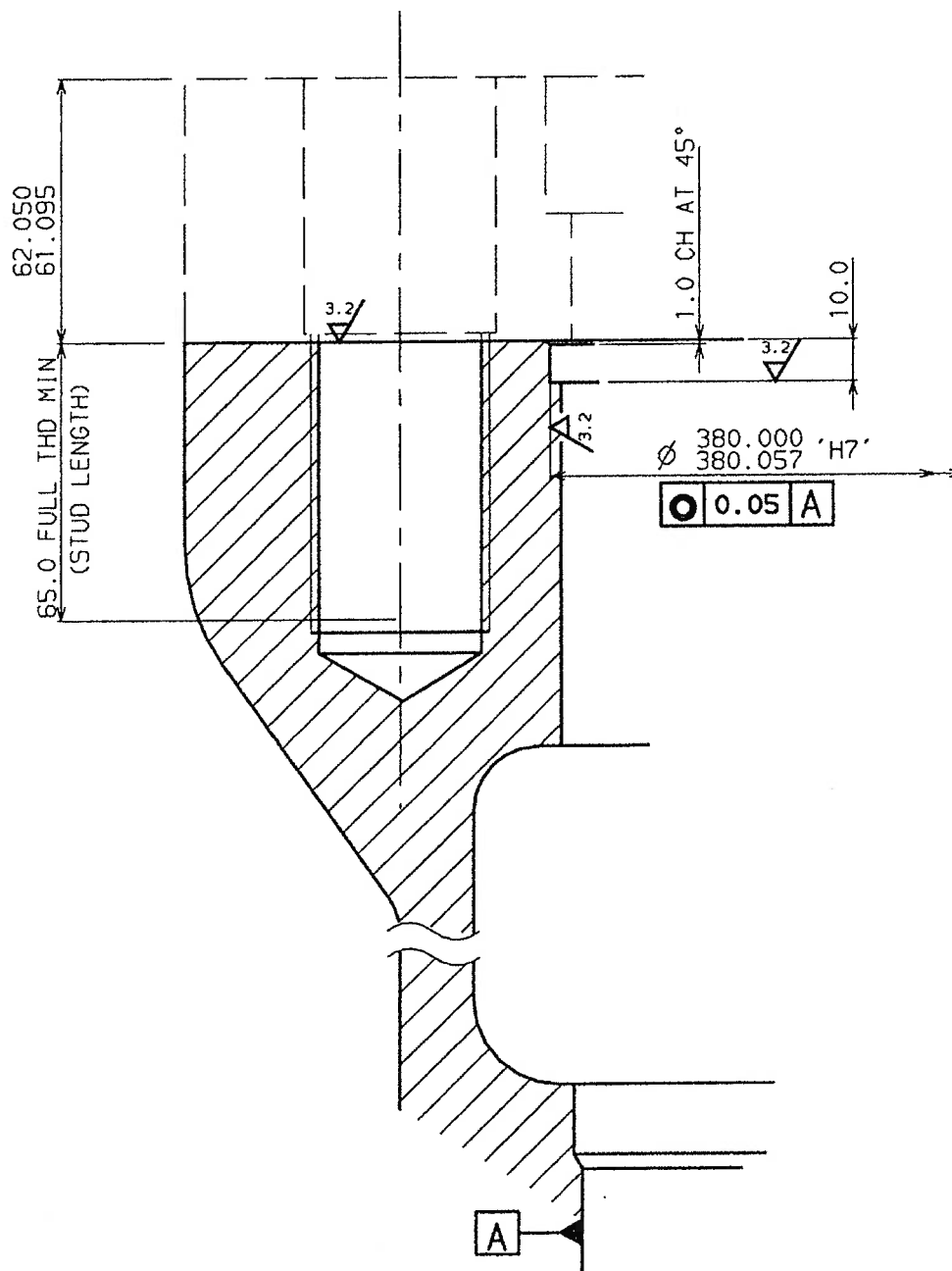


Open water transfer hole from 12 mm diameter to 23.000 to 23.025mm diameter. (aiming for a 0.025mm clearance of sleeve in hole).

Manufacture sleeve from stainless steel as above with grooves near top and bottom 3mm wide x 0.3mm deep for Loctite to collect.

Fit sleeve to transfer hole using loctite 270.

Figure 7 Long Term Repair - Sketch of proposed top liner land modification removing the top cracked portion which will be replaced with a manufactured insert with an 'o'



ring seal.

Brief Report on Rectification of Water Leakage from **Entablature of MP-1 MPPL Abu Road**

Introduction:

3 Nos of Allen S 37 G series engines were installed at Abu Road under KBPL pipeline project. Later on KBPL product pipeline was converted to MPPL crude oil pipeline.

The details of engine are as follows:

Equipment	MP-1
Engine Serial Number	D6/50161-6
Make	M/s WH ALLEN, UK
Type	8S37G HBC

Brief Description of problem:

On October 2011, external water leakage was observed between engine block and Cylinder head no. 6, 7 and 8. Since there was no internal leakage in engine, it was then decided to address the issue during major maintenance which was due in November, 2011.

During the 9000hrs schedule maintenance (CRH: 51807), detailed inspection was carried out and cracks were found in the entablature (in all cylinders) at liner collar landing area. The matter was deliberated with M/s Allen Diesel and old water transfer sleeve were replaced with new water transfer sleeves.

In November 2015 (CRH: 67290) i.e. after running of 15,843 Hrs, external water leakage was again observed from block and Cylinder head no. 8. New sleeves were fabricated of extended length and fitted in Cylinder #8 of engine. After replacement of sleeves, engine trial run was taken and no water leakage was observed.

Since water leakage increased and major maintenance of MP-1 was due, 12000 Hrs maintenance was carried out from 12th May to 7th July 2017 at CRH: 72858. M/s Rai Prexim, Gurgaon was also lined up for inspection of Cylinder block for cracks, checking condition of water transfer sleeves etc. MPI of engine block at liner landing area was carried out which revealed similar circumferential cracks which were reported in 2015. These cracks might have propagated this time further downward. Dimensional inspection of water transfer sleeves for crack and Ovality checking also carried out. Thereafter, sleeves were re-fitted and trial run was taken.

During the load trial on 07th July 2017, external water leakage was again observed from cylinder head and block of cylinder no. 3,4,7 & 8. No water leakage observed from sleeves however leakage observed from liner landing area.

Trouble shooting of the problem:

The problem was further analysed and consultation was done with OEM i.e. M/s Allen Diesel regarding probable cause of water leakage and its best possible solutions. The OEM suggested following long term/ permanent solutions:

1. Machine 62 mm from the top of the cylinder to remove all the cracks and fit top liner inserts. This repair was carried out on MP2 at Sendra in 2013. Approximate expenditure in this case would be Rs 1.0 -1.2 crore and time taken for restoration would be approximately 30 days. Lead time for procurement of material and award of service contract would be approximately 8 months.
2. Replace the cylinder with a new flat top design block. Approximate expenditure in this case would be Rs 3.0-3.5 crore and time taken for restoration would be approximately 40 days. Lead time for procurement of material and award of service contract would be approximately 12 months.

Since both the options were time consuming and attending the water leakage problem would have been possible only after 8 to 10 months, a short terms solution was conceptualised. Approximate expenditure in this case would be Rs 5.0-6.0 Lakh and time taken for restoration would be approximately 8-10 days. Lead time for procurement of material and award of service contract would be approximately 1 month.

The short term solution involves replacement of existing water transfer sleeve with new sleeves of longer length and groove for O-ring which will act as additional sealing between water transfer sleeve and hole. Sleeves were manufactured from stainless steel (SS-312) and fitted back using Loctite 270. This additional arrangement of O-ring is supposed to block of leakage path around the water transfer holes and arresting the leakage of water.

Since there was also possibility of crack propagation to underneath the top portion of block to water jacket, water seepage to liner landing area through this path may occur. To prevent this external leakage, modification in steel ring was done. Viton O-ring was fitted in outer dimension of steel ring by making small groove so that water can be stopped from coming out of block.

The preventive maintenance for arresting of water leakage by way of water transfer sleeve replacement was carried out from 25.08.2017 to 08.09.2017. The removal of already fitted water transfer sleeves without damaging the water transfer hole in the block was very challenging job. The sleeves were removed by jacking arrangement (as shown in figure-1). As the sleeves were fitted using "Araldite", pull force on the sleeves were very high due to which the stud being used for jacking arrangement sheared frequently. The broken studs were welded with another stud and again the jacking was done to remove the sleeves.

After removal of all the sleeves, the water transfer holes were cleaned thoroughly. Dimension of all water transfer holes were also measured for any ovality. New water transfer sleeves with enhanced length and o-ring were fitted by using Loctite 270. After fitting of all water transfer sleeves, modified steel rings were fitted and the engine was boxed up. Water filling of the engine was done and no water leakage was observed. The load trial run of the MP-1 was taken on 08.09.2017. The engine is running & all the operational parameters are well within limit.



Figure 1 : Removal of sleeves by jacking



Figure 2: Broken stud being used for jacking



Figure 3: Removal of sleeves by jacking

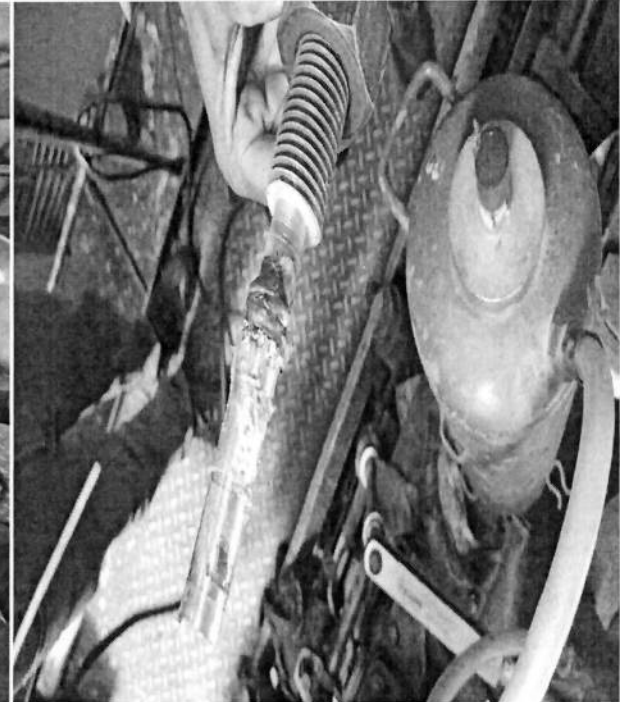


Figure 4: Sleeve with welded stud



Figure 5: Safety precautions before welding

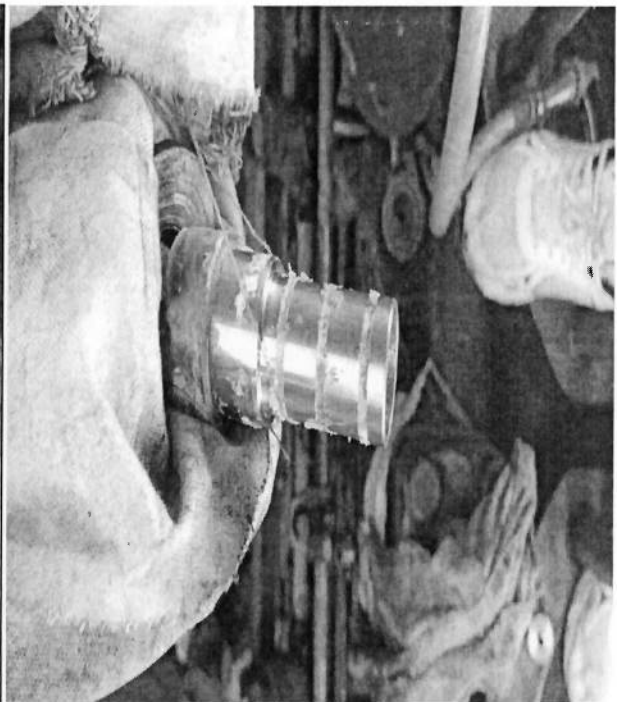


Figure 6: Sleeve removed through jacking



Figure 7: Sleeves retrieved from engine block



Figure 8: water transfer hole after sleeve removal



Figure 9: Water transfer sleeve with O-ring



Figure 10: Water transfer sleeve with O-ring

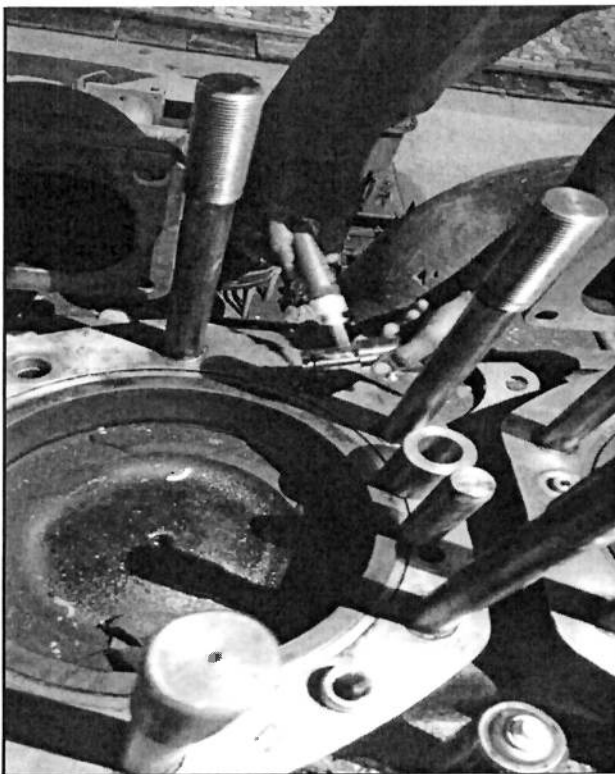


Figure 11: Fitting of water transfer sleeve

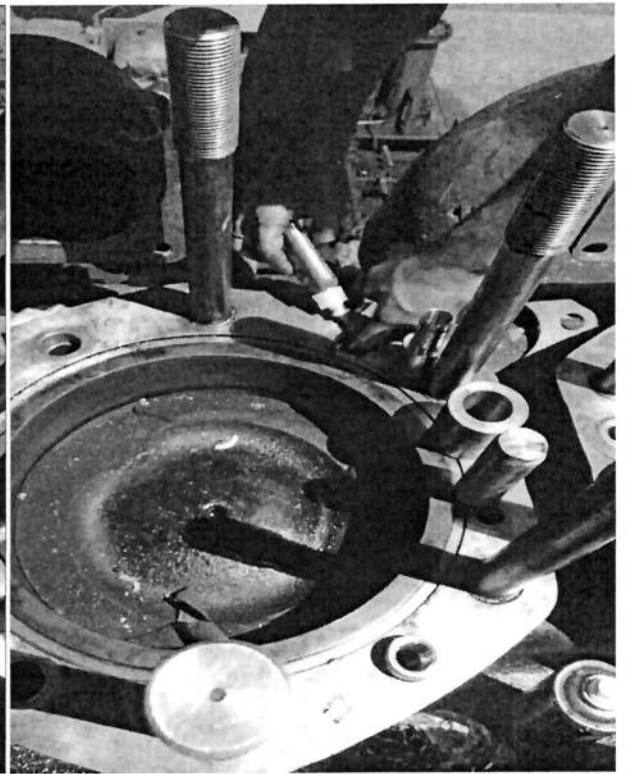


Figure 12: Fitting of water transfer sleeve

